# Intercambiadores de calor

## Sesión 3 de 3 · Efectividad y NUT

Gestión Energética II · ILN222

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Universidad Técnica Federico Santa María · 2026-2

Çengel y Ghajar, cap. 11

# Plan de la clase

## La unidad

1. Dos fluidos y una pared
2. Diferencia de temperatura media logarítmica
3. Efectividad y NUT

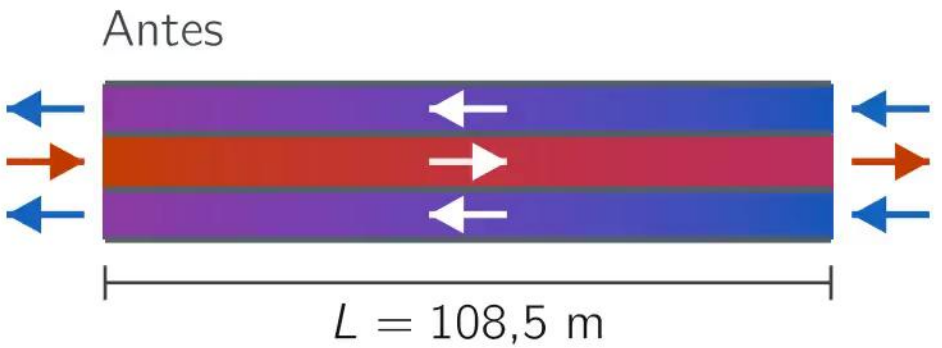
## Hoy

- **Evaluar un equipo**, el equipo existe y se buscan las salidas
- **Calor máximo y efectividad**, contraflujo infinito y $\varepsilon$
- **Curvas $\varepsilon$ contra NUT**, casos límite y el colector solar
- **Ejemplos y selección**, el ejemplo geotérmico con $\varepsilon$ y NUT

# ¿El doble de calor?

Pregunta de control

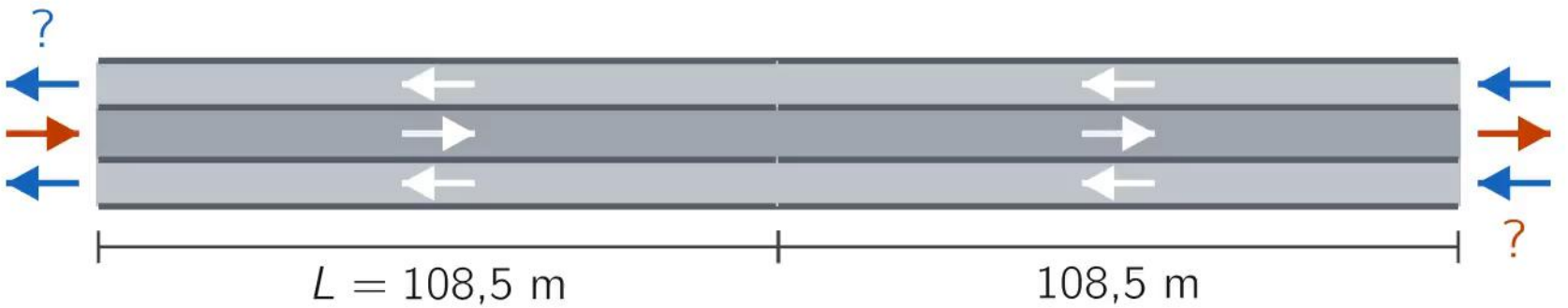
Si duplico el intercambiador, ¿duplico el calor?



# ¿El doble de calor?

Pregunta de control

Si duplico el intercambiador, ¿duplico el calor?



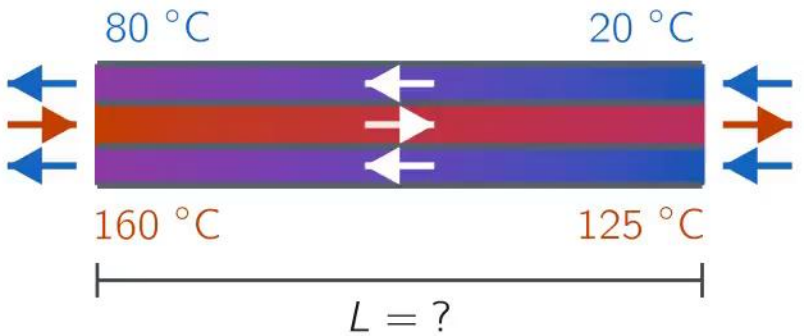

☐ Sí ☐ Menos del doble ☐ Más del doble

# Evaluar un equipo existente

$\Delta T_1$ extremo 1 $\Delta T_2$ extremo 2 $\Delta T_{ml}$ media logarítmica
$U$ total $A_s$ área $\dot{Q}$ calor

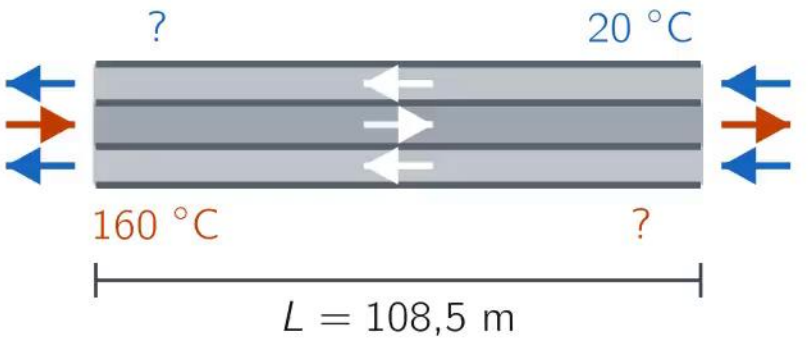
## Dimensionar

Se conocen las salidas, se busca el tamaño




## Evaluar

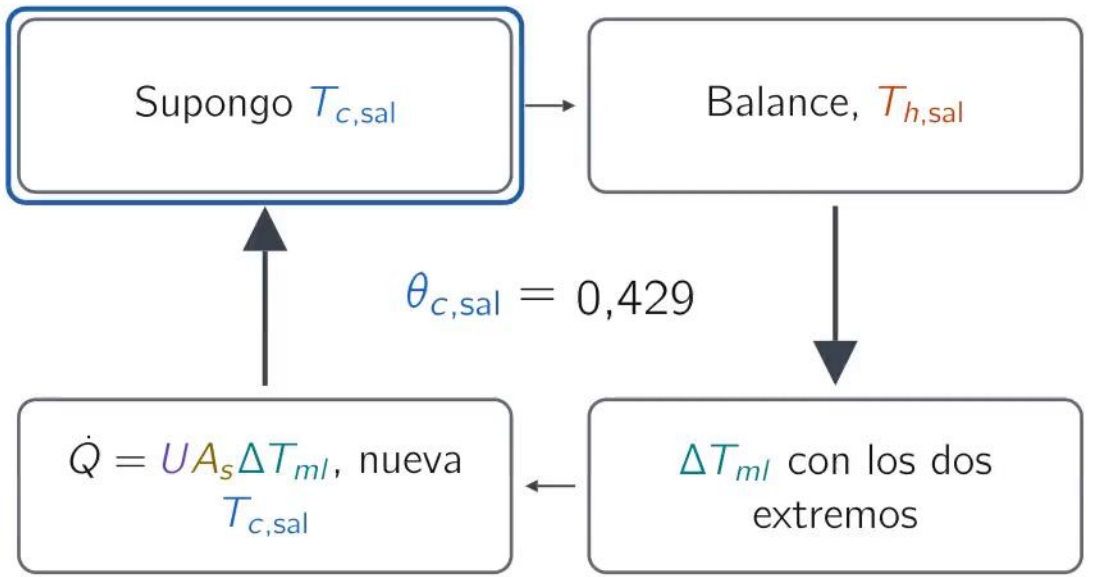
Se conoce el equipo, se buscan las salidas



# Evaluar un equipo existente

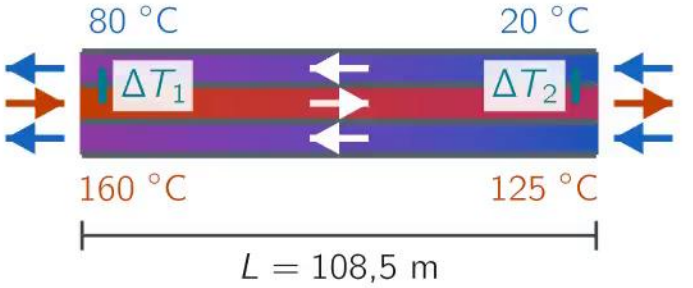
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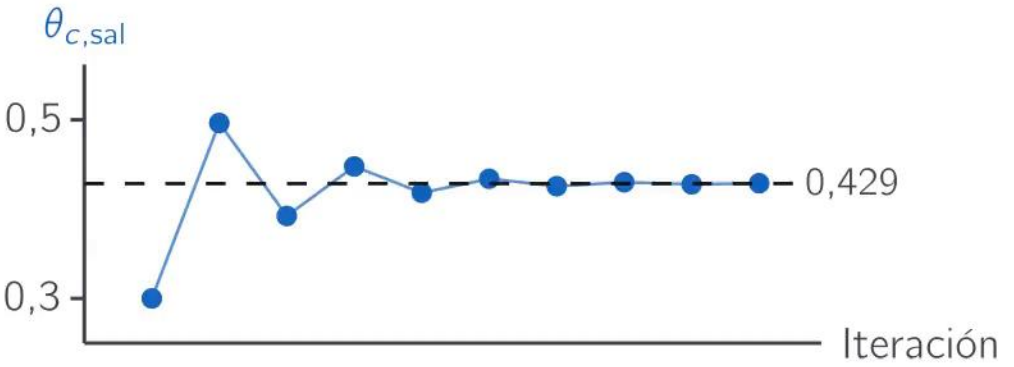
$$\Delta T_{ml} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}$$




**Evaluar**

Se conoce el equipo, se buscan las salidas



# Contenidos


1 Evaluar un equipo

2 Calor máximo y efectividad

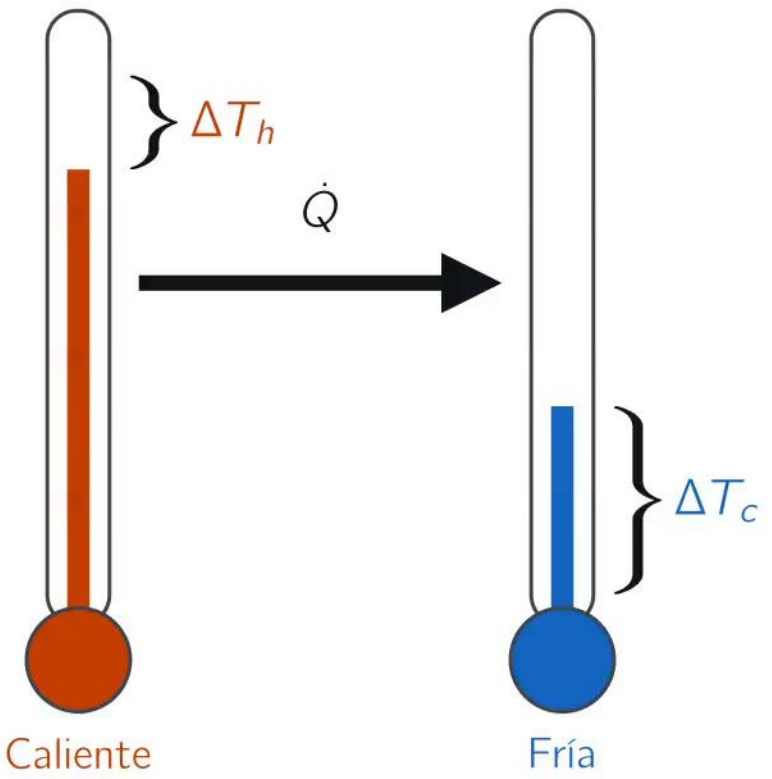
3 Curvas $\varepsilon$–NUT

4 Ejemplos y selección

# Capacidad calorífica

$C_h$ caliente $C_c$ fría $c = C_{\min}/C_{\max}$

$C_{\min}$ menor $C_{\max}$ mayor $\Delta T_{\max} = T_{h,\text{ent}} - T_{c,\text{ent}}$




$$\dot{Q} = C\,\Delta T, \qquad C = \dot{m}\,c_p$$

Capacidad calorífica, W/°C (kW/°C en los ejemplos)

$$C_h = \dot{m}_h\,c_{p,h}, \qquad C_c = \dot{m}_c\,c_{p,c}$$

$$C_{\min} = C_c, \qquad C_{\max} = C_h$$

$$c = C_{\min}/C_{\max}$$

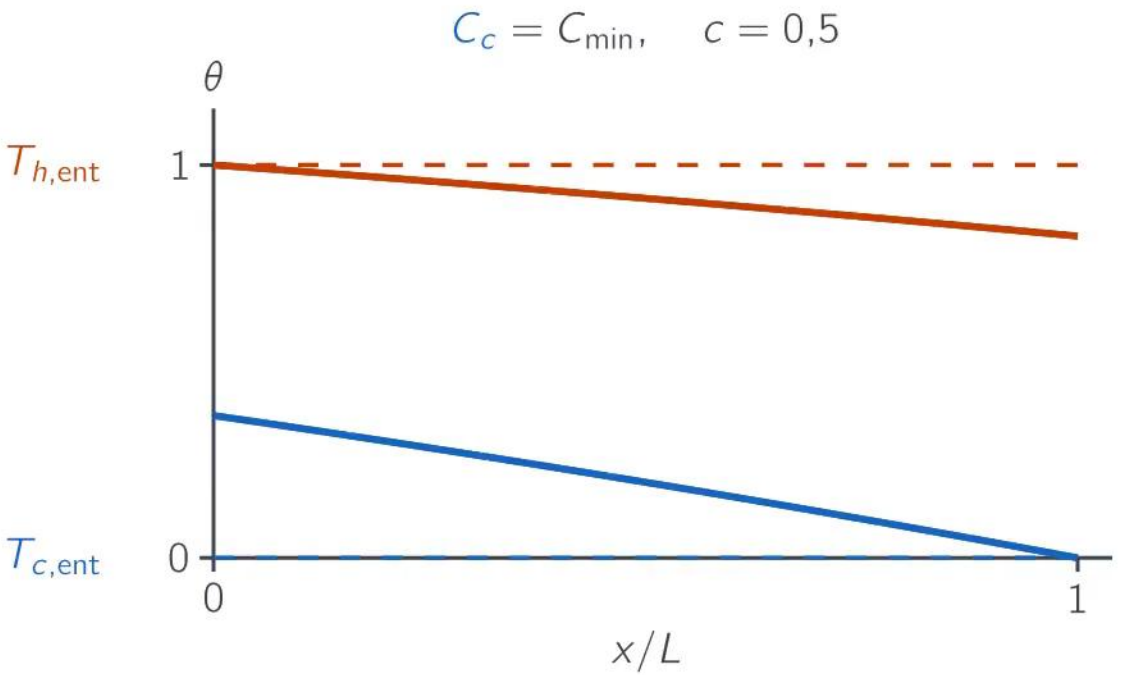
No confundir con $c_p$

# Capacidad calorífica

$C_h$ caliente $C_c$ fría $c = C_{\min}/C_{\max}$

$C_{\min}$ menor $C_{\max}$ mayor $\Delta T_{\max} = T_{h,\text{ent}} - T_{c,\text{ent}}$

$$\dot{Q} = C_c\,(T_{c,\text{sal}} - T_{c,\text{ent}})$$
$$= C_h\,(T_{h,\text{ent}} - T_{h,\text{sal}})$$



# Capacidad calorífica

$C_h$ caliente $C_c$ fría $c = C_{\min}/C_{\max}$

$C_{\min}$ menor $C_{\max}$ mayor $\Delta T_{\max} = T_{h,\text{ent}} - T_{c,\text{ent}}$

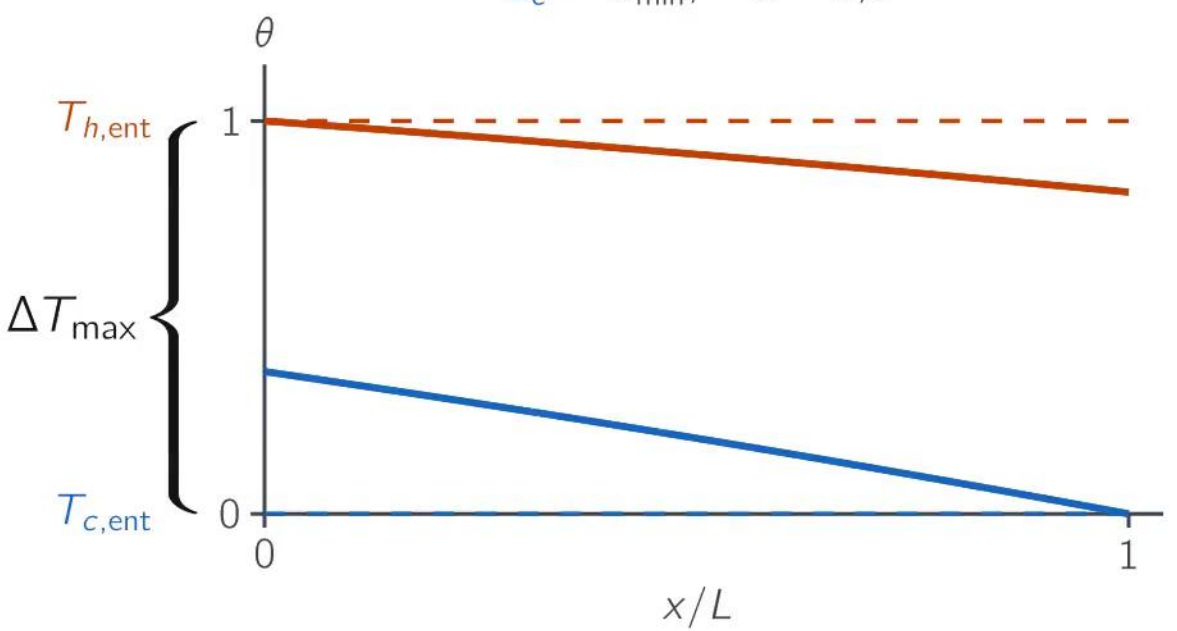
Pregunta de control

¿Puede algún fluido cambiar su temperatura más que $\Delta T_{\max}$?

$$\dot{Q} = C_c\,(T_{c,\text{sal}} - T_{c,\text{ent}}) = C_h\,(T_{h,\text{ent}} - T_{h,\text{sal}})$$

$$\Delta T_{\max} = T_{h,\text{ent}} - T_{c,\text{ent}}$$

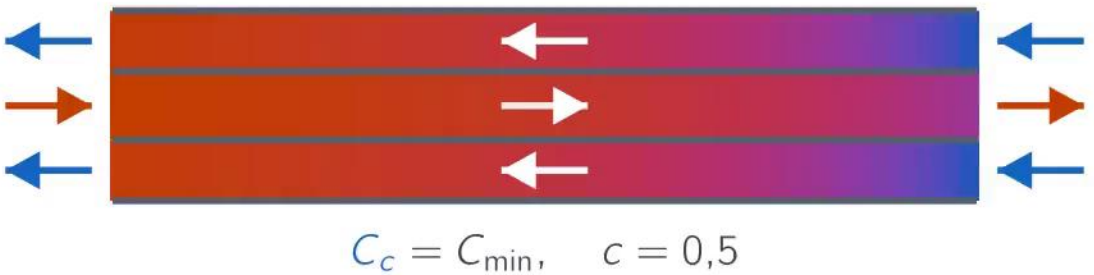
$C_c = C_{\min}, \quad c = 0{,}5$



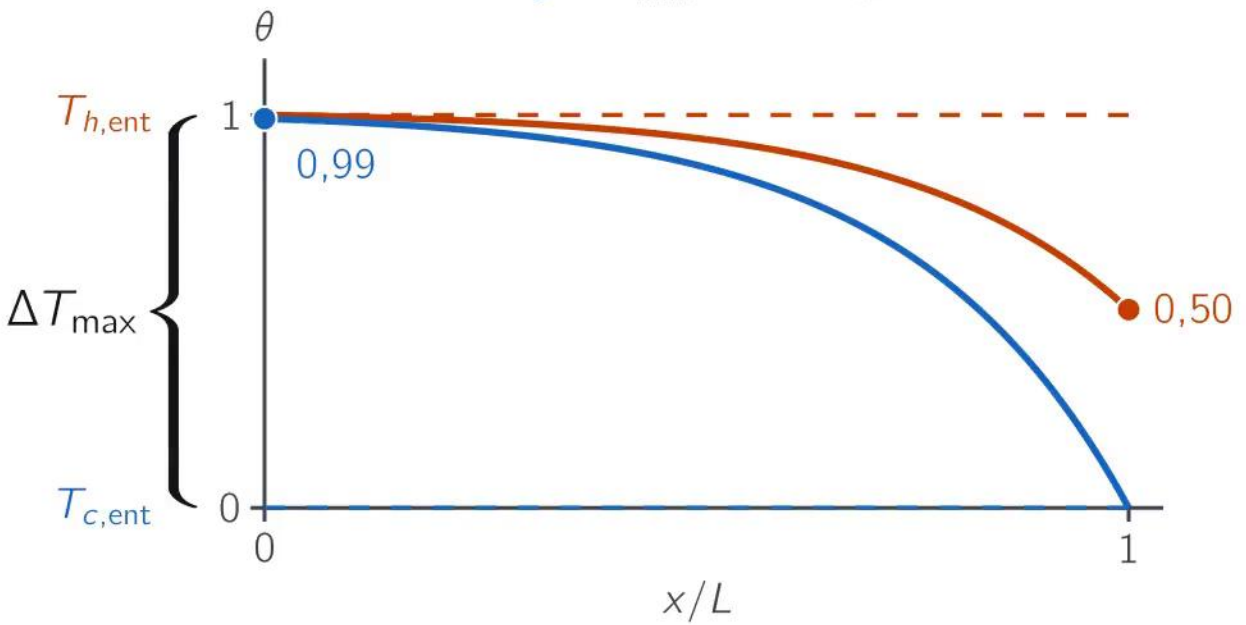
# Calor máximo

$C_h$ caliente $C_c$ fría $c = C_{\min}/C_{\max}$

$C_{\min}$ menor $C_{\max}$ mayor $\Delta T_{\max} = T_{h,\text{ent}} - T_{c,\text{ent}}$



$C_c = C_{\min}, \quad c = 0{,}5$




Caso límite

$\dot{Q} \le C_c \, \Delta T_{\max}$

$\dot{Q} \le C_h \, \Delta T_{\max}$

$\Rightarrow \quad \dot{Q} \le C_{\min} \, \Delta T_{\max}$

$$\dot{Q}_{\max} = C_{\min} \, \Delta T_{\max}$$

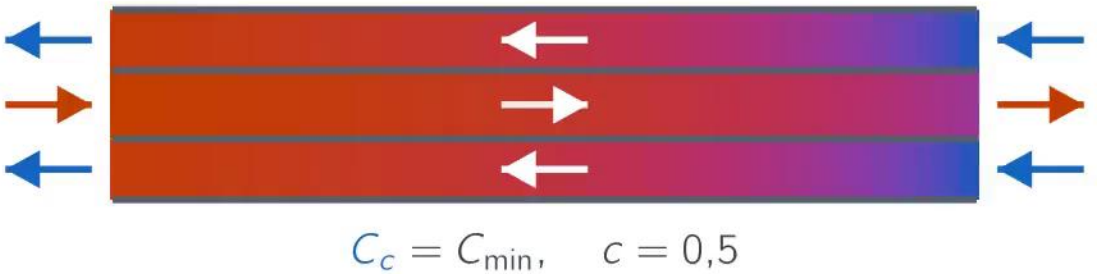



$L_0$ largo de partida

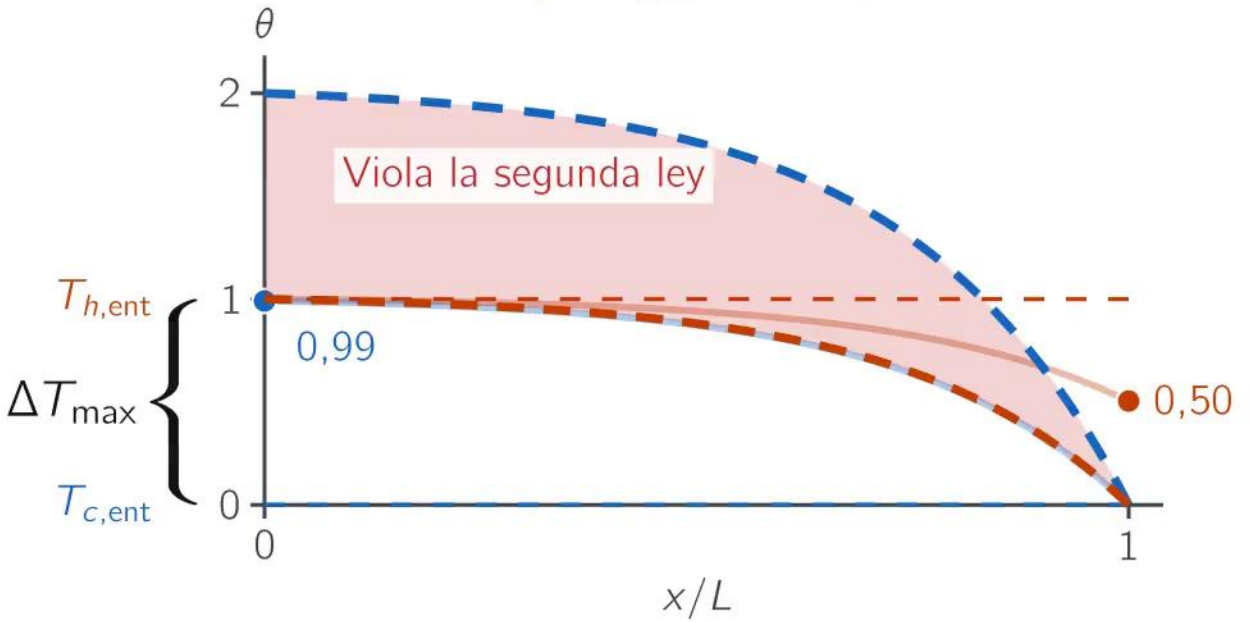
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$C_h$ caliente $C_c$ fría $c = C_{\min}/C_{\max}$

$C_{\min}$ menor $C_{\max}$ mayor $\Delta T_{\max} = T_{h,\text{ent}} - T_{c,\text{ent}}$



$C_c = C_{\min}, \quad c = 0{,}5$




Error frecuente

$$\dot{Q} \le C_c\,\Delta T_{\max}$$

$$\dot{Q} \le C_h\,\Delta T_{\max}$$

$$\Rightarrow\ \dot{Q} \le C_{\min}\,\Delta T_{\max}$$

$$\boxed{\dot{Q}_{\max} = C_{\min}\,\Delta T_{\max}}$$

~~$\dot{Q}_{\max} = C_{\max}\,\Delta T_{\max}$ ?~~

$$\Rightarrow\ \Delta T_c = \frac{\dot{Q}}{C_{\min}} = \frac{\Delta T_{\max}}{c}$$

$$= 2\,\Delta T_{\max} > \Delta T_{\max}$$

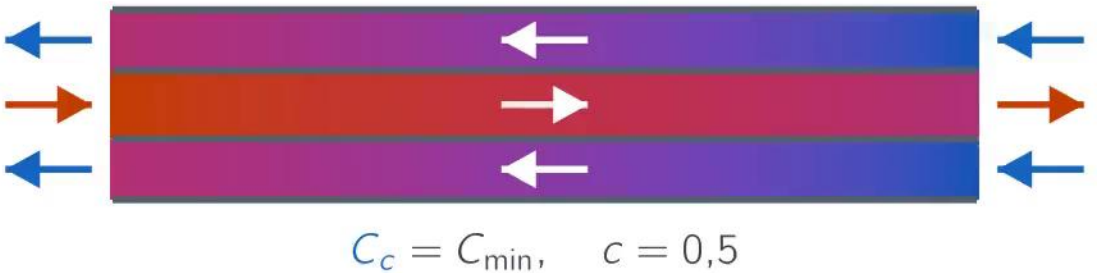


# Efectividad

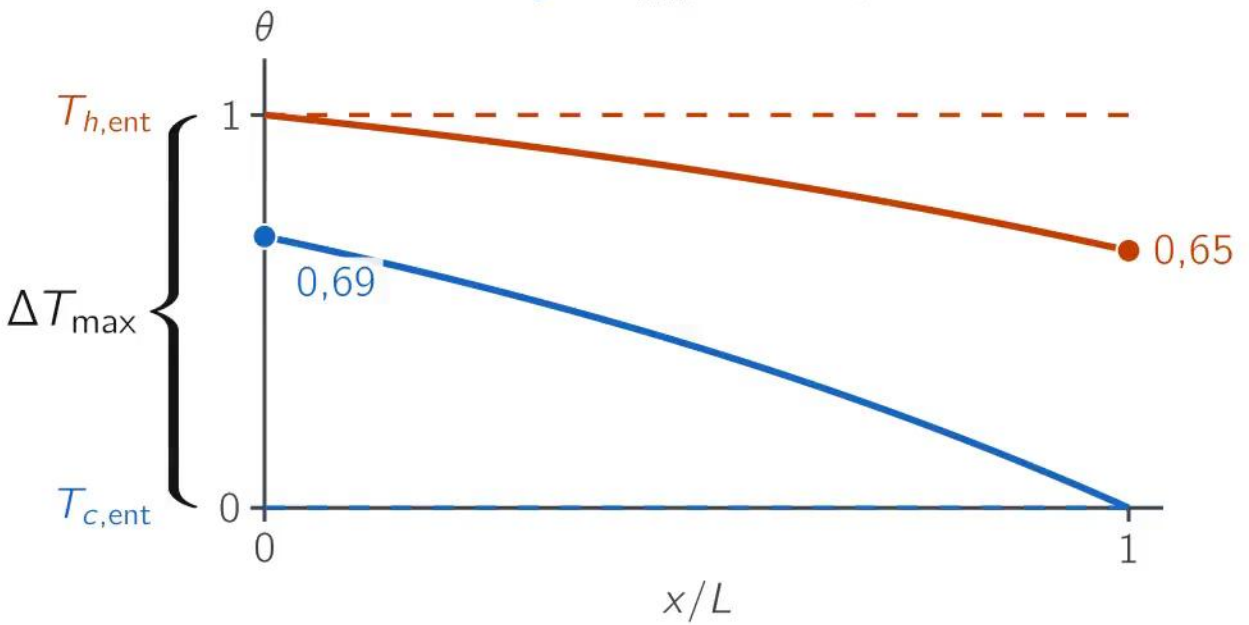
$\varepsilon$ efectividad  NUT unidades de transferencia  $c$ razón

$\dot{Q}_{max}$ máximo  $C_{min}$ menor  $UA_s$ conductancia

$$\dot{Q}_{max} = C_{min}\,\Delta T_{max}$$

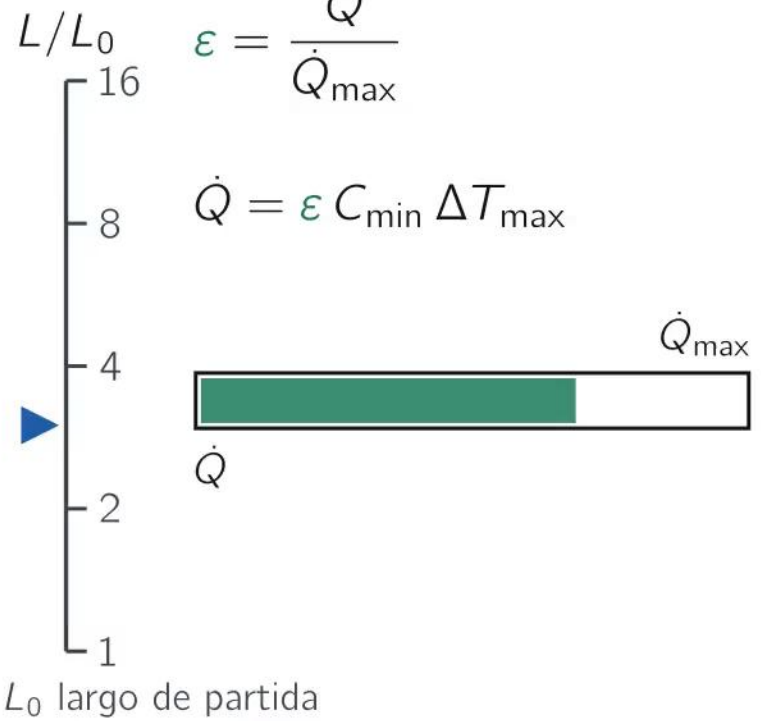


$C_c = C_{min}, \quad c = 0{,}5$




$$\varepsilon = \frac{\dot{Q}}{\dot{Q}_{max}}$$

$$\dot{Q} = \varepsilon\, C_{min}\,\Delta T_{max}$$

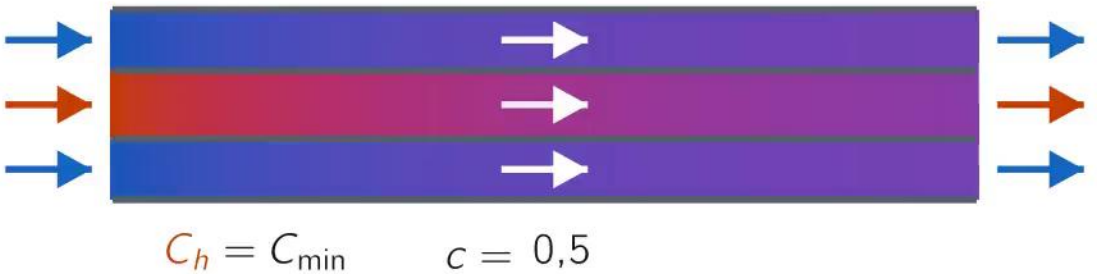


# Efectividad

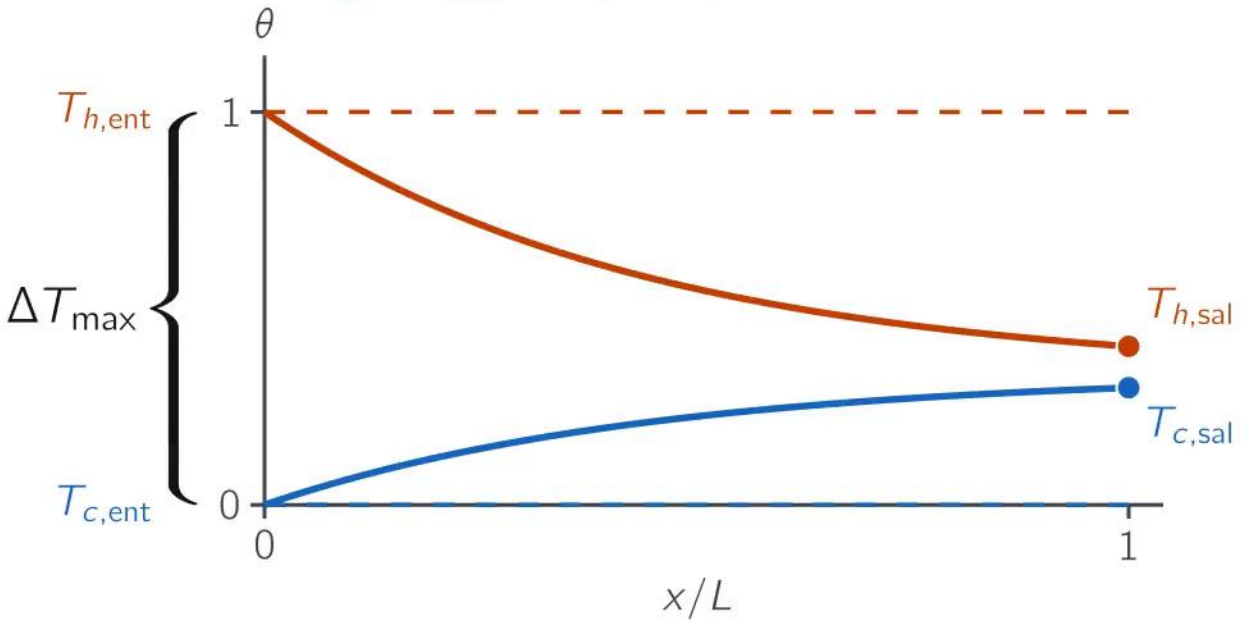
$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia

$$\dot{Q} = \varepsilon\, C_{min}\, \Delta T_{max}$$

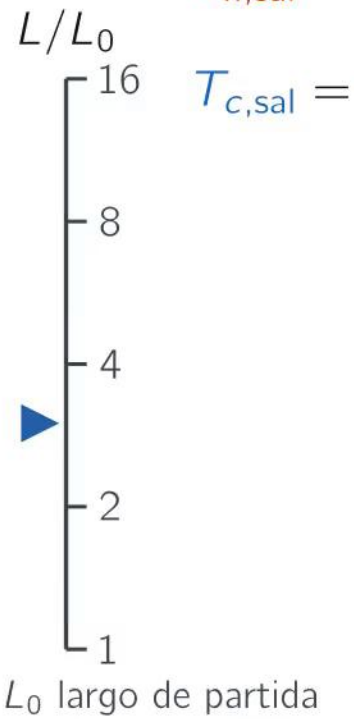


$C_h = C_{min}$ $c = 0{,}5$




$$T_{h,sal} = T_{h,ent} - \varepsilon\, \Delta T_{max}$$

$$T_{c,sal} = T_{c,ent} + \varepsilon\, c\, \Delta T_{max}$$



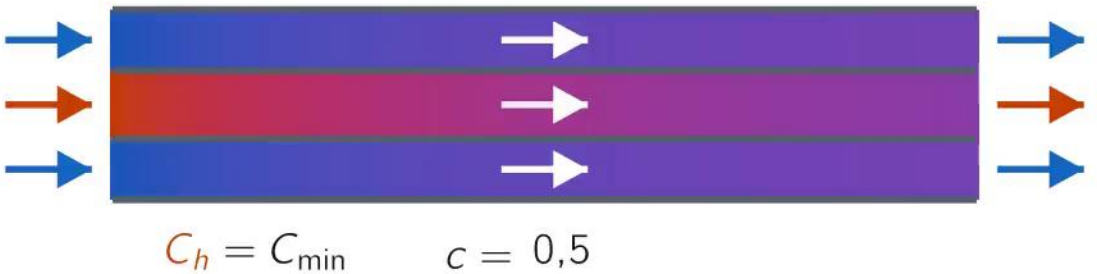

$L_0$ largo de partida

# Efectividad

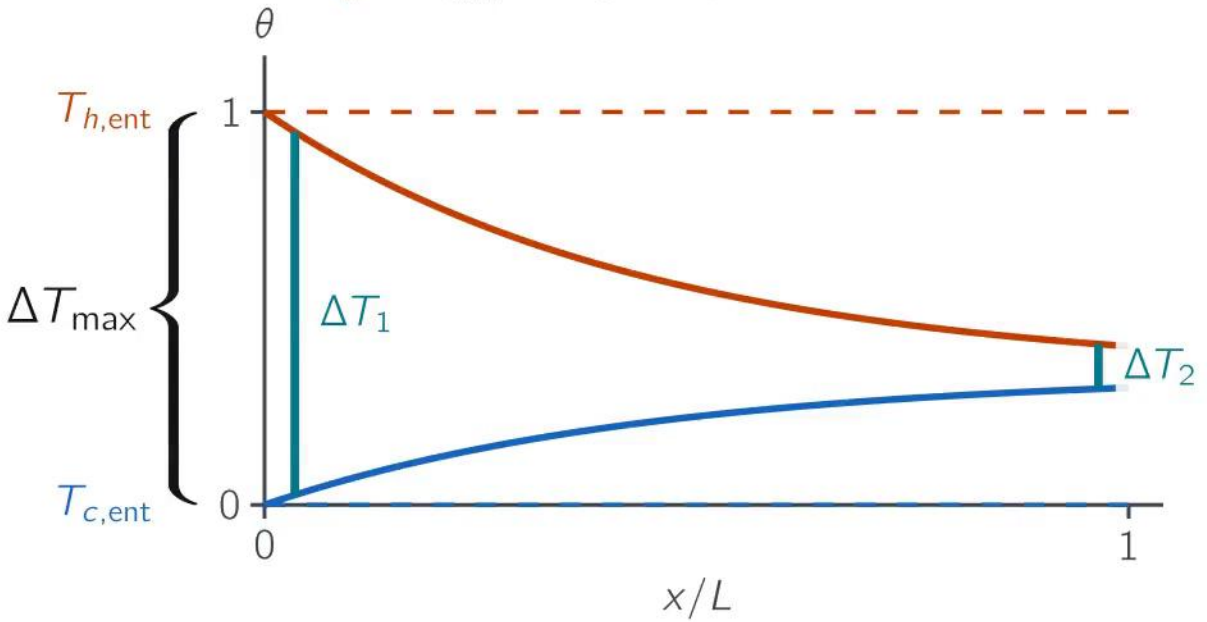
$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia

$$\dot{Q} = \varepsilon\, C_{min}\, \Delta T_{max}$$



$C_h = C_{min}$ $c = 0{,}5$

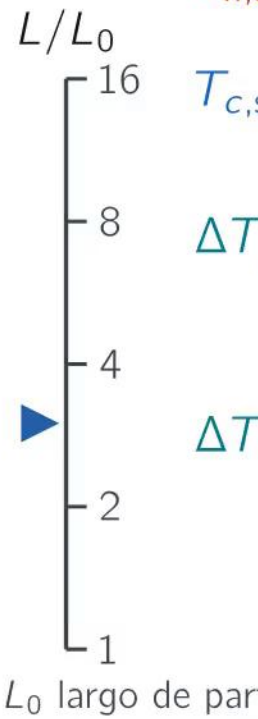



$$T_{h,sal} = T_{h,ent} - \varepsilon\, \Delta T_{max}$$

$$T_{c,sal} = T_{c,ent} + \varepsilon\, c\, \Delta T_{max}$$

$$\Delta T_1 = T_{h,ent} - T_{c,ent} = \Delta T_{max}$$

$$\Delta T_2 = T_{h,sal} - T_{c,sal} = \Delta T_{max}\,[1 - \varepsilon(1 + c)]$$

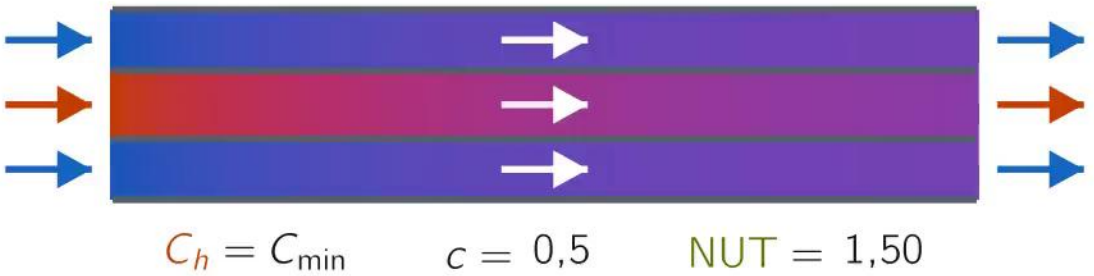



$L_0$ largo de partida

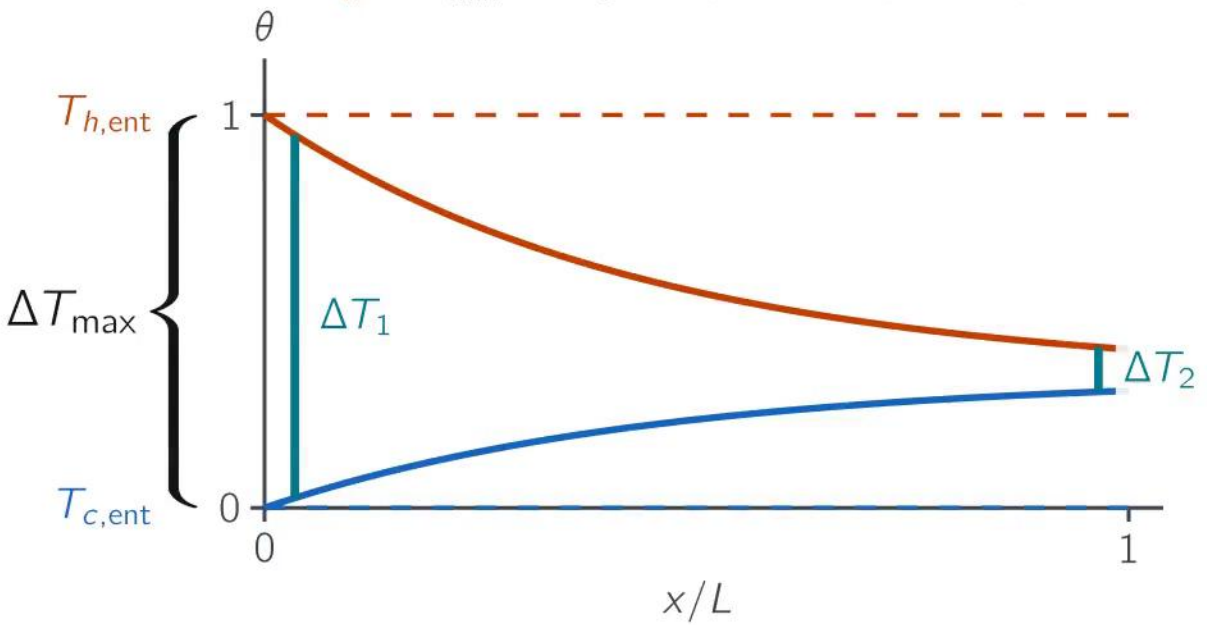
# Efectividad

$\varepsilon$ efectividad  NUT unidades de transferencia  $c$ razón

$\dot{Q}_{\max}$ máximo  $C_{\min}$ menor  $UA_s$ conductancia



$C_h = C_{\min}$  $c = 0{,}5$  NUT $= 1{,}50$




$$d\dot{Q} = U\, dA\, \Delta T$$

$$d\Delta T = -d\dot{Q}\left(\frac{1}{C_h} + \frac{1}{C_c}\right)$$

Se integra del extremo 1 al 2

$$\ln\frac{\Delta T_2}{\Delta T_1} = -\frac{UA_s}{C_{\min}}(1+c)$$

$$\frac{1}{C_h} + \frac{1}{C_c} = \frac{1}{C_{\min}}\left(1 + \frac{C_{\min}}{C_{\max}}\right) = \frac{1+c}{C_{\min}}$$

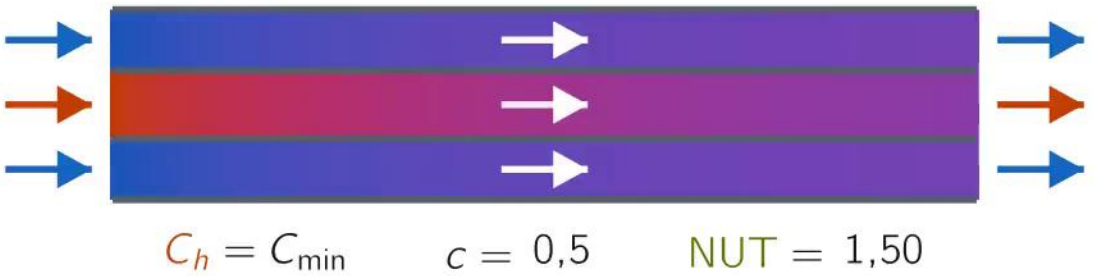
$$\text{NUT} = \frac{UA_s}{C_{\min}}$$

# Efectividad

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

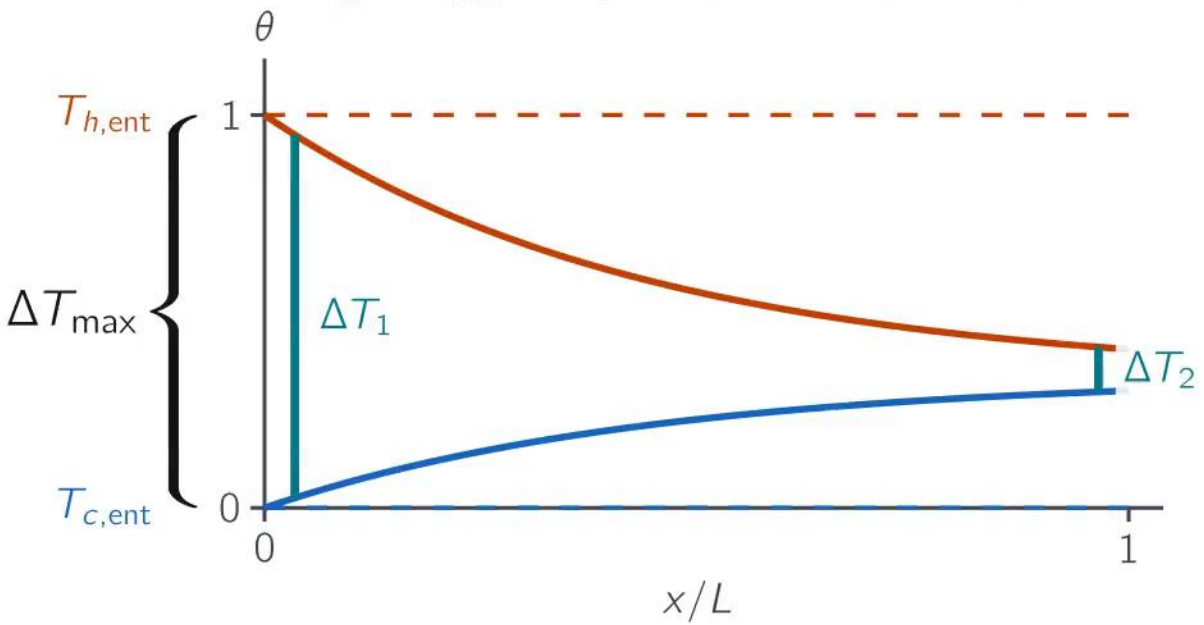
$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia

$$\ln \frac{\Delta T_2}{\Delta T_1} = -\text{NUT}(1+c) \qquad \text{NUT} = \frac{UA_s}{C_{min}}$$



$C_h = C_{min}$ $c = 0{,}5$ NUT $= 1{,}50$

$$\frac{\Delta T_2}{\Delta T_1} = \frac{\Delta T_{max}\,[1-\varepsilon(1+c)]}{\Delta T_{max}} = 1-\varepsilon(1+c)$$



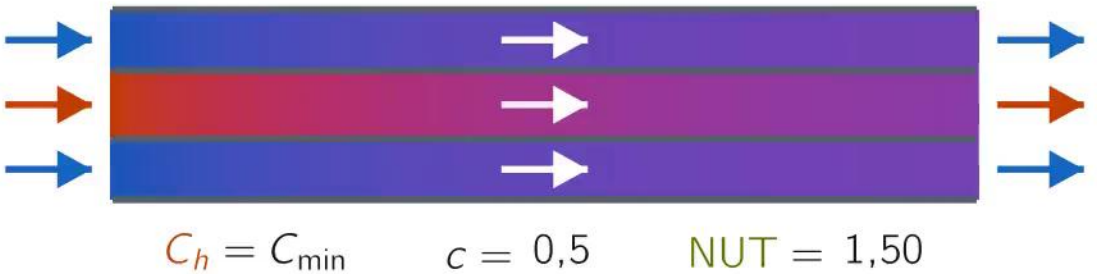

$$\ln[1-\varepsilon(1+c)] = -\text{NUT}(1+c)$$

$$\Delta T_1 = \Delta T_{max}$$
$$\Delta T_2 = \Delta T_{max}\,[1-\varepsilon(1+c)]$$

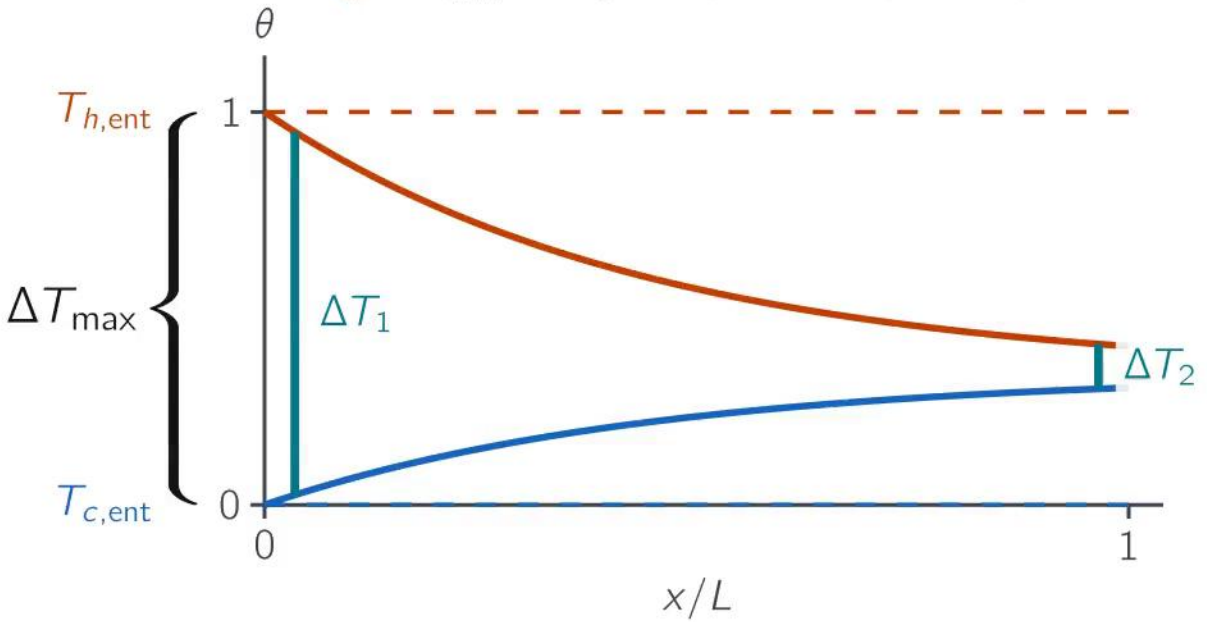
# Efectividad

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia



$C_h = C_{min}$ $c = 0{,}5$ NUT $= 1{,}50$




$$\ln \frac{\Delta T_2}{\Delta T_1} = -\text{NUT}(1+c) \qquad \text{NUT} = \frac{UA_s}{C_{min}}$$

$$\frac{\Delta T_2}{\Delta T_1} = \frac{\Delta T_{max}\,[1-\varepsilon(1+c)]}{\Delta T_{max}} = 1-\varepsilon(1+c)$$

$$\ln[1-\varepsilon(1+c)] = -\text{NUT}(1+c)$$

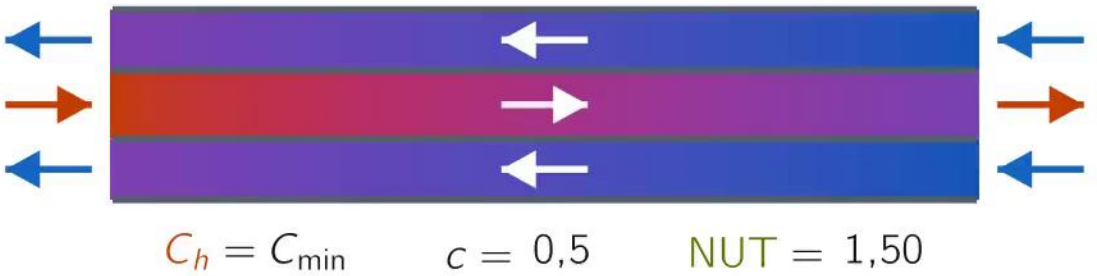
$$1-\varepsilon(1+c) = e^{-\text{NUT}(1+c)}$$

$$\boxed{\varepsilon = \frac{1-e^{-\text{NUT}(1+c)}}{1+c}}$$ Paralelo

# Efectividad

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia



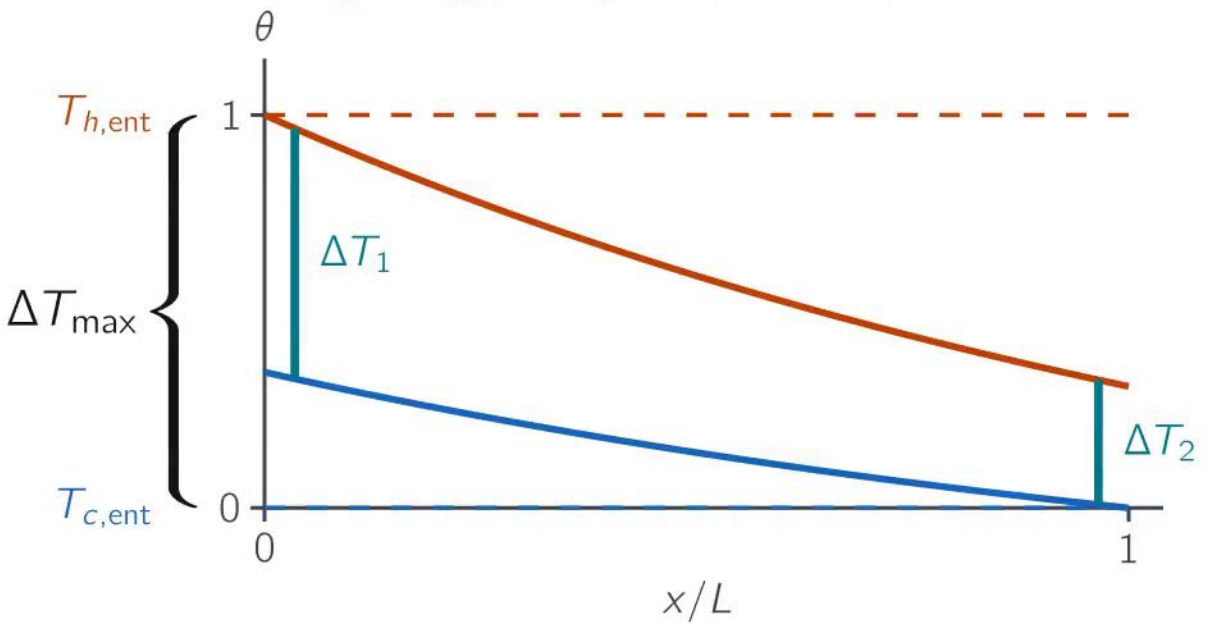
$C_h = C_{min}$ $c = 0{,}5$ NUT $= 1{,}50$

$$\varepsilon = \frac{1 - e^{-\text{NUT}(1+c)}}{1+c}$$ Paralelo $$\text{NUT} = \frac{UA_s}{C_{min}}$$

$$d\Delta T = -d\dot{Q}\left(\frac{1}{C_h} - \frac{1}{C_c}\right)$$

$$\Delta T_1 = \Delta T_{max}(1 - \varepsilon c)$$

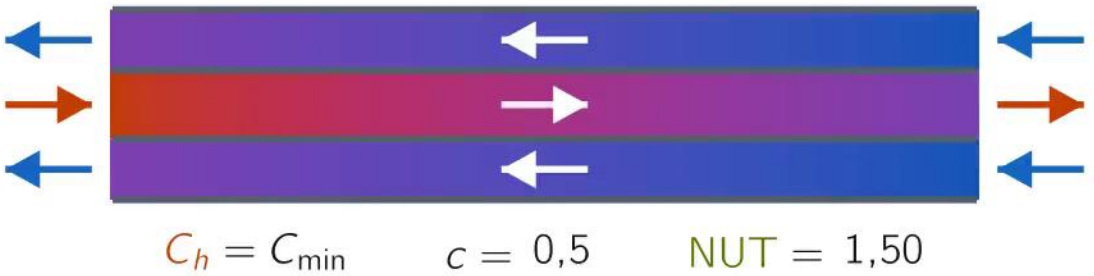
$$\Delta T_2 = \Delta T_{max}(1 - \varepsilon)$$



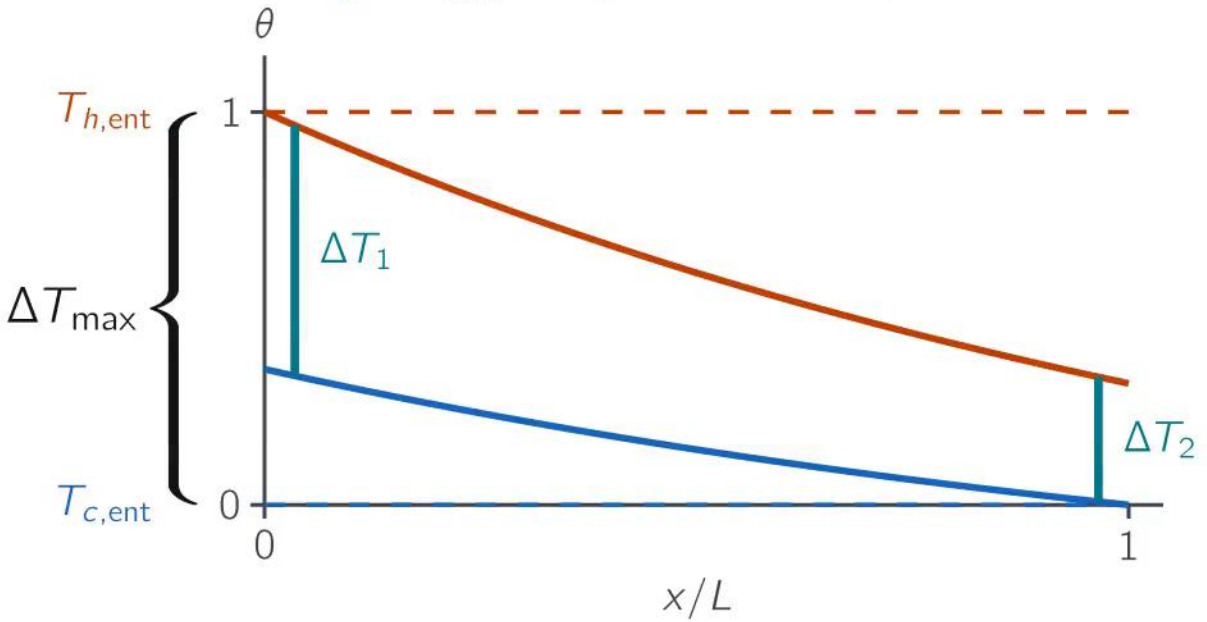
# Efectividad

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia



$C_h = C_{min}$ $c = 0{,}5$ NUT $= 1{,}50$




$$\varepsilon = \frac{1 - e^{-\mathrm{NUT}(1+c)}}{1+c}$$ Paralelo $\mathrm{NUT} = \frac{UA_s}{C_{min}}$

$$\mathrm{d}\Delta T = -\mathrm{d}\dot{Q}\left(\frac{1}{C_h} - \frac{1}{C_c}\right)$$

$$\Delta T_1 = \Delta T_{max}\,(1 - \varepsilon c)$$

$$\Delta T_2 = \Delta T_{max}\,(1 - \varepsilon)$$

$$\ln\frac{1-\varepsilon}{1-\varepsilon c} = -\mathrm{NUT}\,(1-c)$$

$$1 - \varepsilon = e^{-\mathrm{NUT}(1-c)}\,(1 - \varepsilon c)$$

$$\varepsilon = \frac{1 - e^{-\mathrm{NUT}(1-c)}}{1 - c\,e^{-\mathrm{NUT}(1-c)}}$$ Contraflujo

# Efectividad

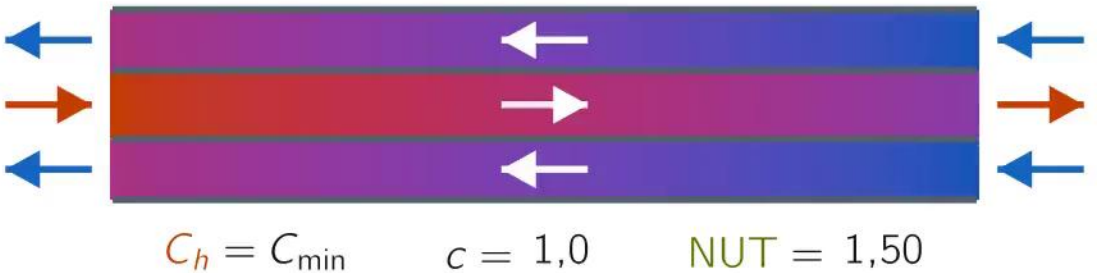
$\varepsilon$ efectividad  NUT unidades de transferencia  $c$ razón

$\dot{Q}_{max}$ máximo  $C_{min}$ menor  $UA_s$ conductancia

$$\varepsilon = \frac{1 - e^{-\mathrm{NUT}(1+c)}}{1+c}$$ Paralelo $$\mathrm{NUT} = \frac{UA_s}{C_{min}}$$

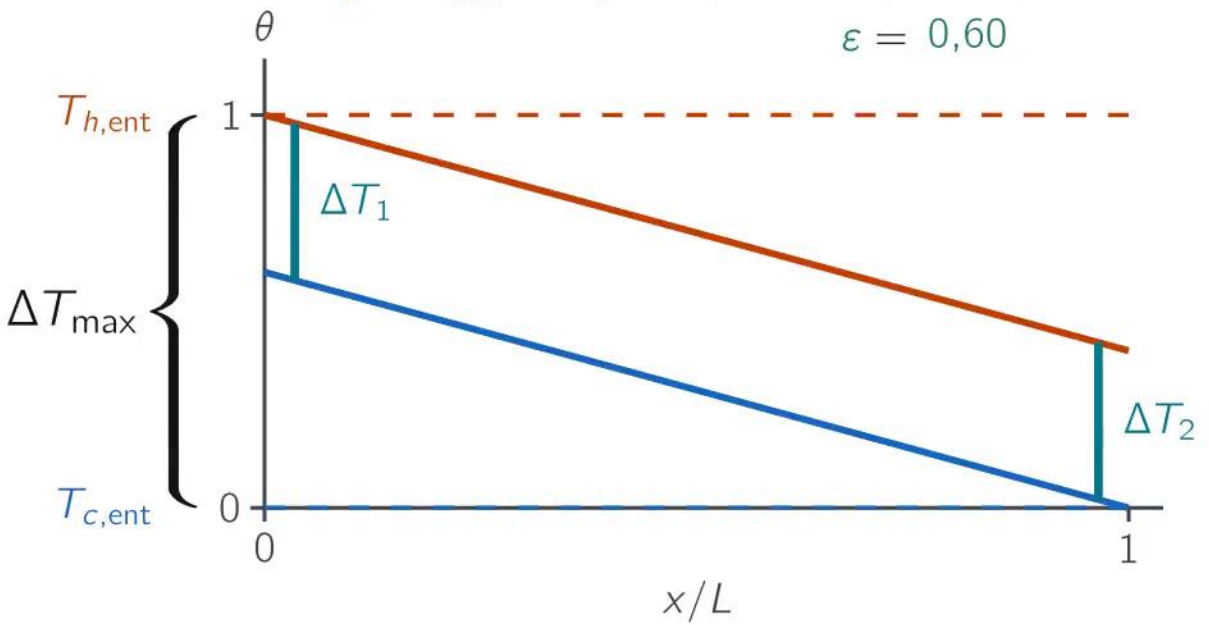
$$\varepsilon = \frac{1 - e^{-\mathrm{NUT}(1-c)}}{1 - c\,e^{-\mathrm{NUT}(1-c)}}$$ Contraflujo

$c = 1$ da 0/0



$C_h = C_{min}$  $c = 1{,}0$  NUT $= 1{,}50$

$\varepsilon = 0{,}60$




Caso límite $$c = 1: \quad \varepsilon = \frac{\mathrm{NUT}}{1+\mathrm{NUT}}$$

# Efectividad

$\varepsilon$ efectividad  NUT unidades de transferencia  $c$ razón

$\dot{Q}_{max}$ máximo  $C_{min}$ menor  $UA_s$ conductancia

$$\varepsilon = f(\text{NUT},\ c,\ \text{configuración})$$

$$\varepsilon = \frac{1 - e^{-\text{NUT}(1+c)}}{1+c}$$

Paralelo

Doble tubo, paralelo y contraflujo

$$\varepsilon = \frac{1 - e^{-\text{NUT}(1-c)}}{1 - c\, e^{-\text{NUT}(1-c)}}$$

Contraflujo

Casco y tubos

$$c = 1: \quad \varepsilon = \frac{\text{NUT}}{1+\text{NUT}}$$

Flujo cruzado, ambos no mezclados

Aproximada

Çengel, tabla 11-4

# Contenidos


1 Evaluar un equipo

2 Calor máximo y efectividad

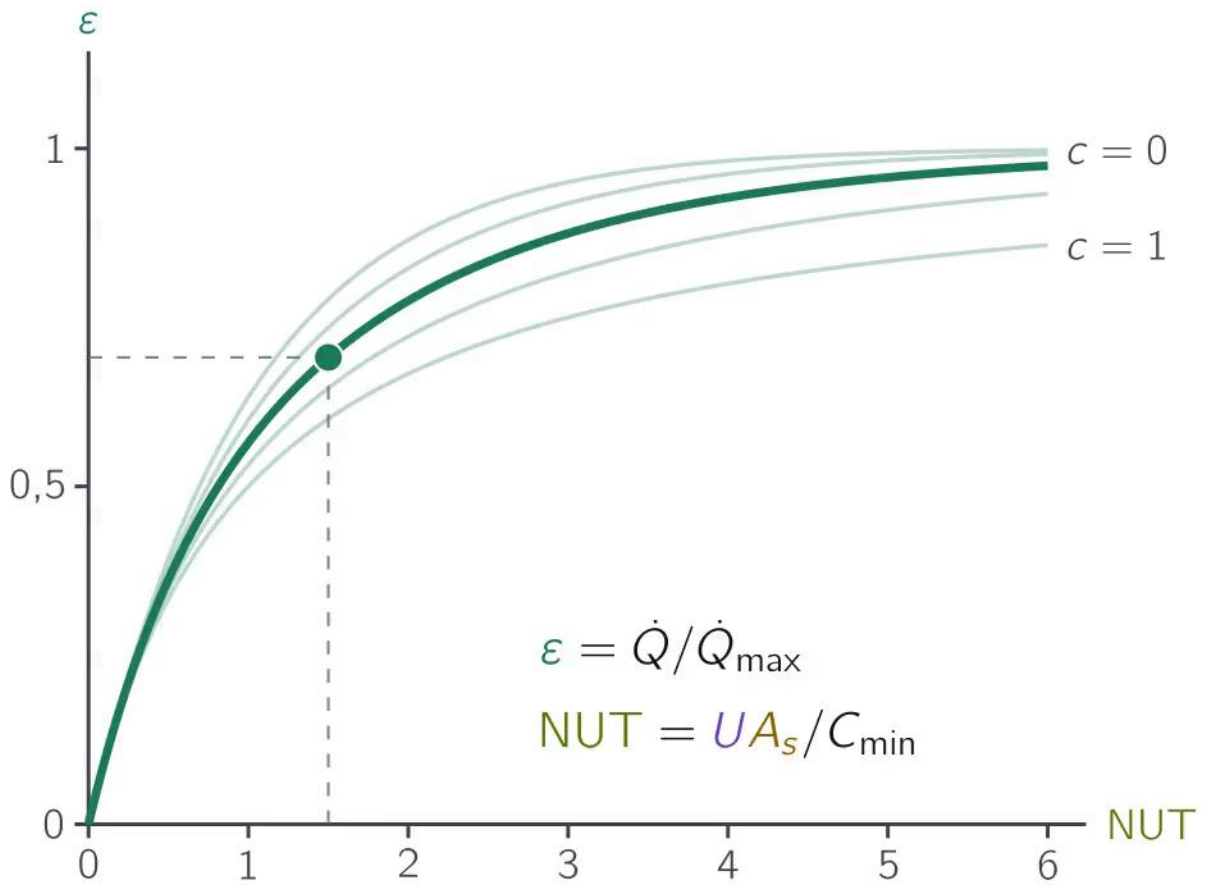
3 Curvas $\varepsilon$–NUT

4 Ejemplos y selección

# Efectividad contra NUT

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia

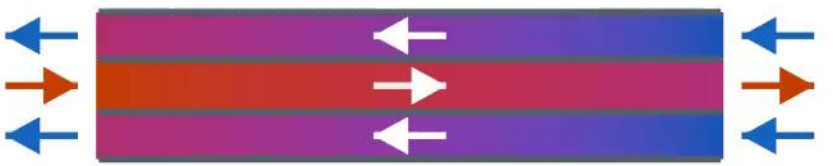


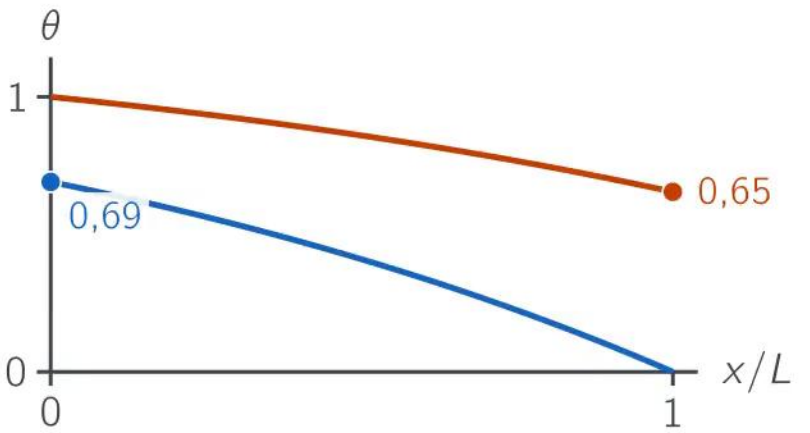

$c = 0{,}50$

$\text{NUT} = 1{,}50$

$\varepsilon = 0{,}69$

$C_{min} = C_c$

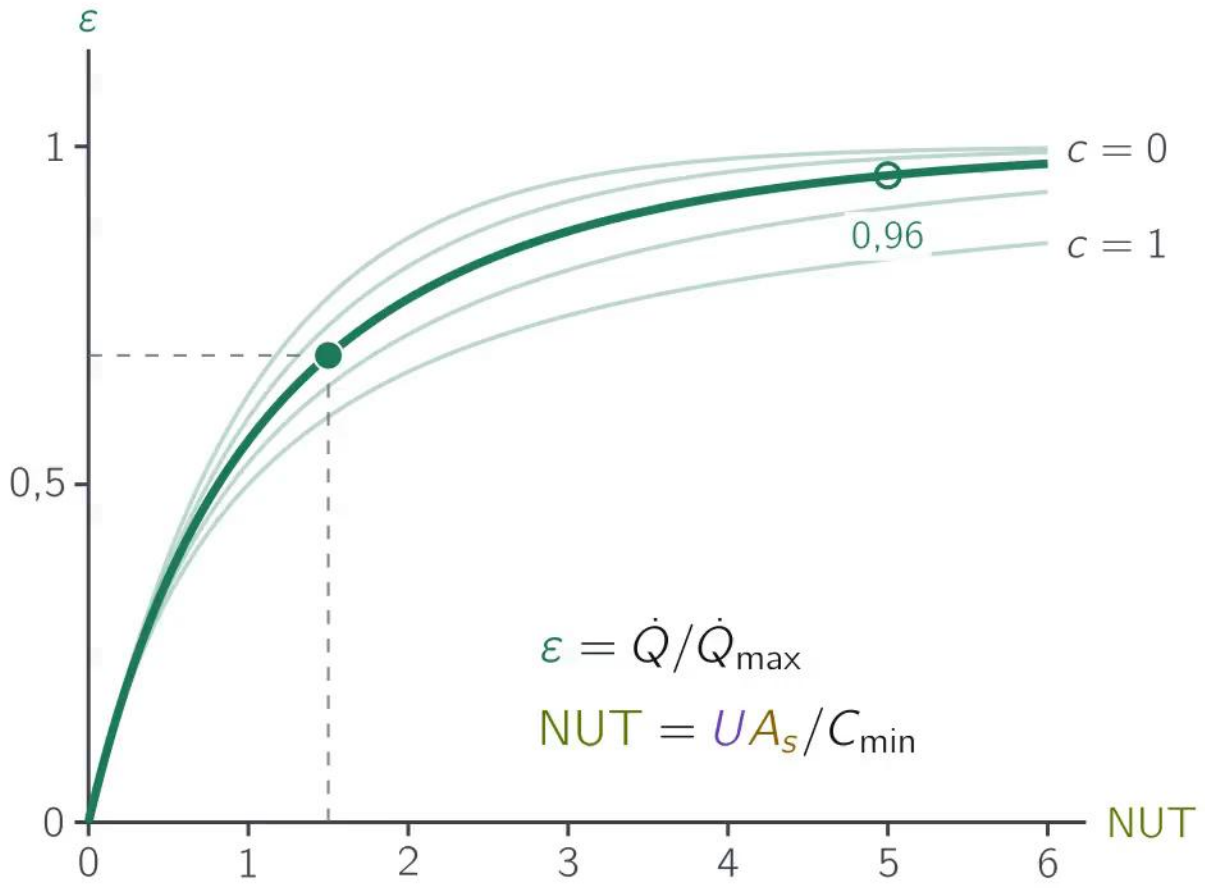




# Efectividad contra NUT

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia

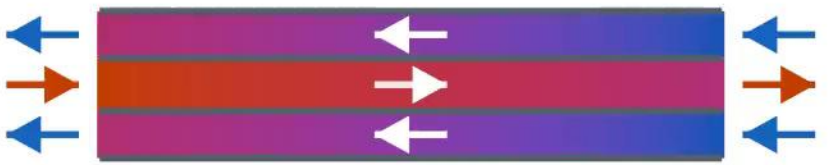


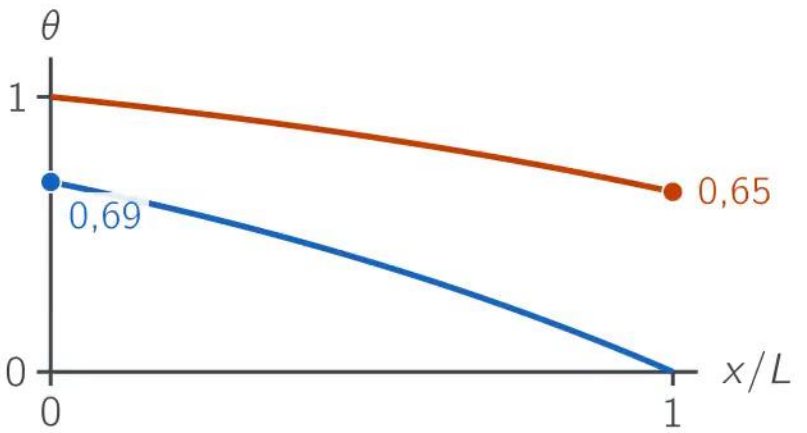

$c = 0{,}50$

$\text{NUT} = 1{,}50$

$\varepsilon = 0{,}69$

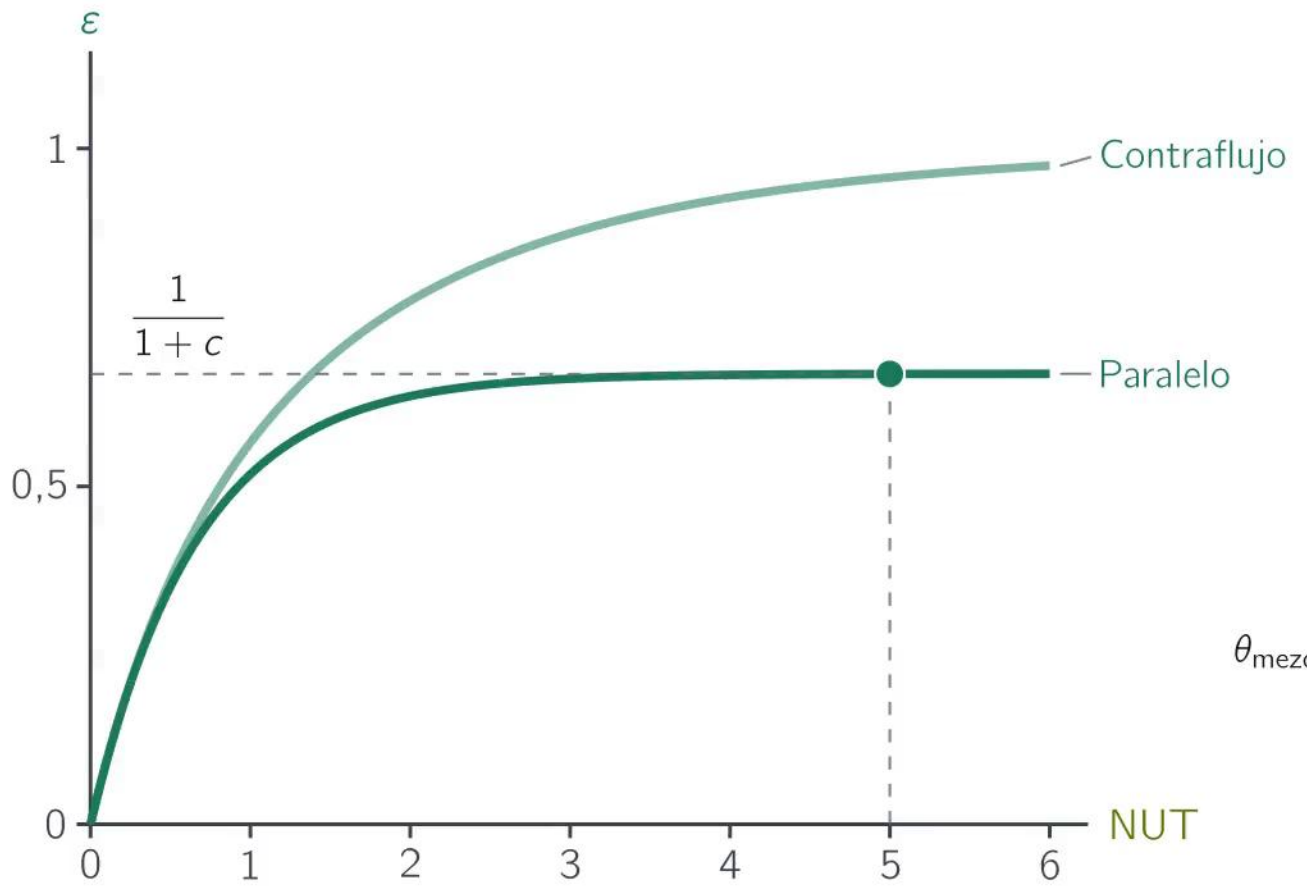
$C_{min} = C_c$





# Efectividad contra NUT

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón
$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia

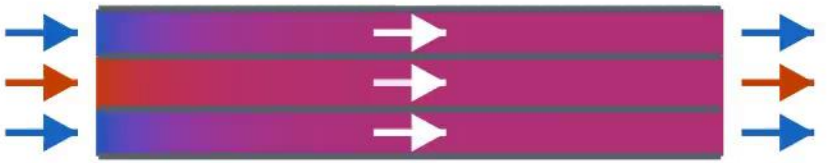


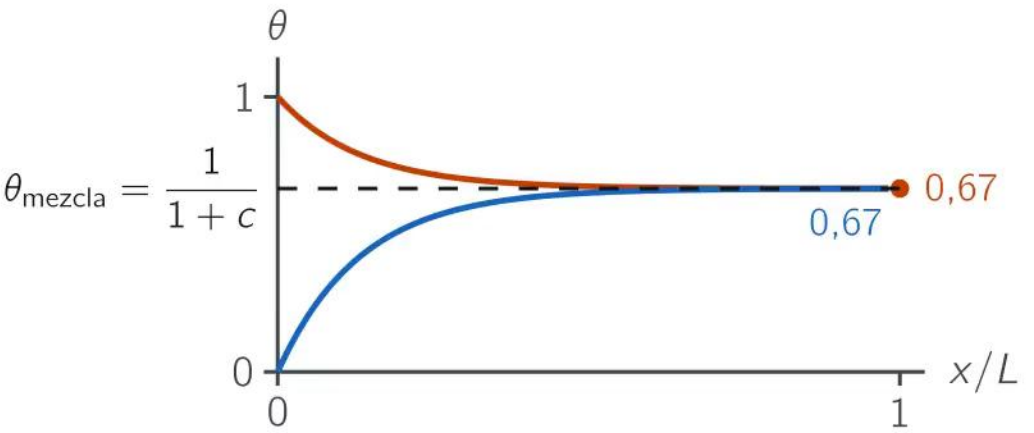

$c = 0{,}50$

$\text{NUT} = 5{,}00$

$\varepsilon = 0{,}67$

$C_{min} = C_c$

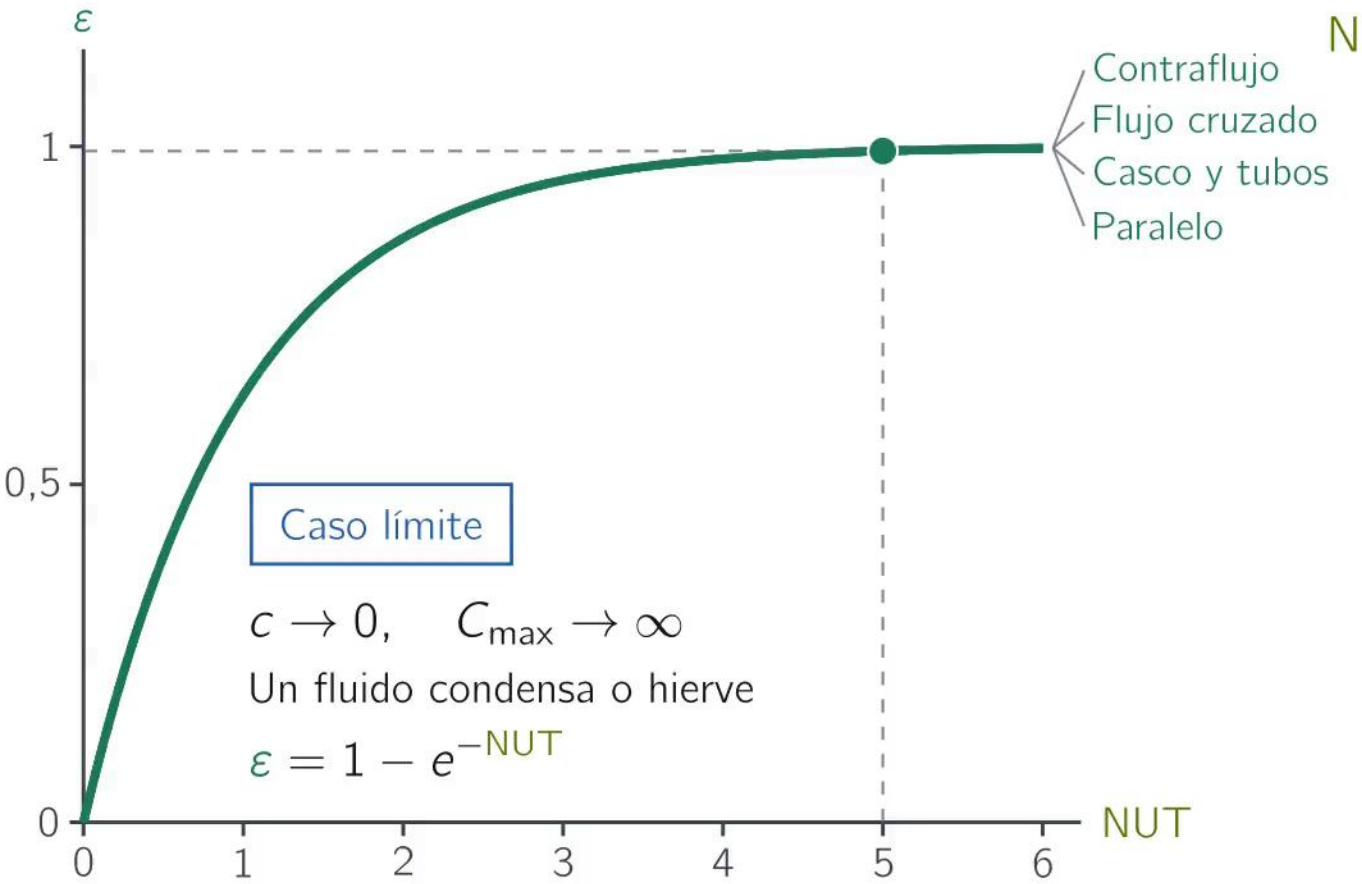




# Efectividad contra NUT

$\varepsilon$ efectividad  NUT unidades de transferencia  $c$ razón

$\dot{Q}_{max}$ máximo  $C_{min}$ menor  $UA_s$ conductancia

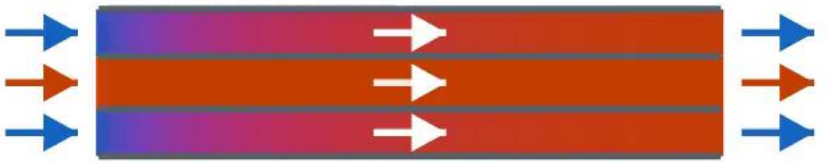


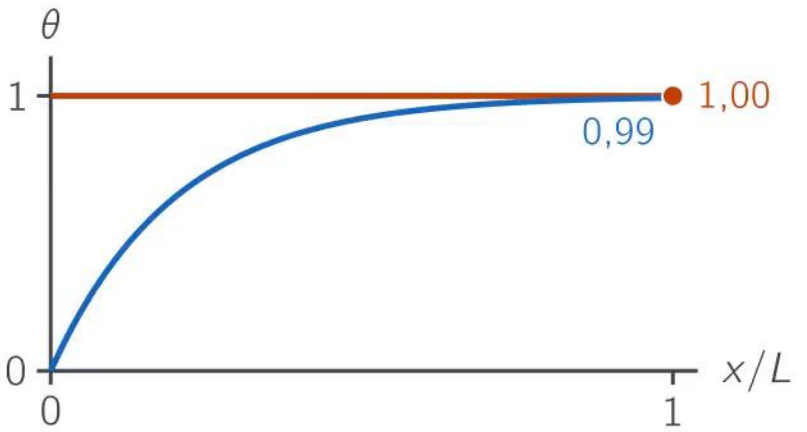

$c = 0{,}00$

$NUT = 5{,}00$

$\varepsilon = 0{,}99$

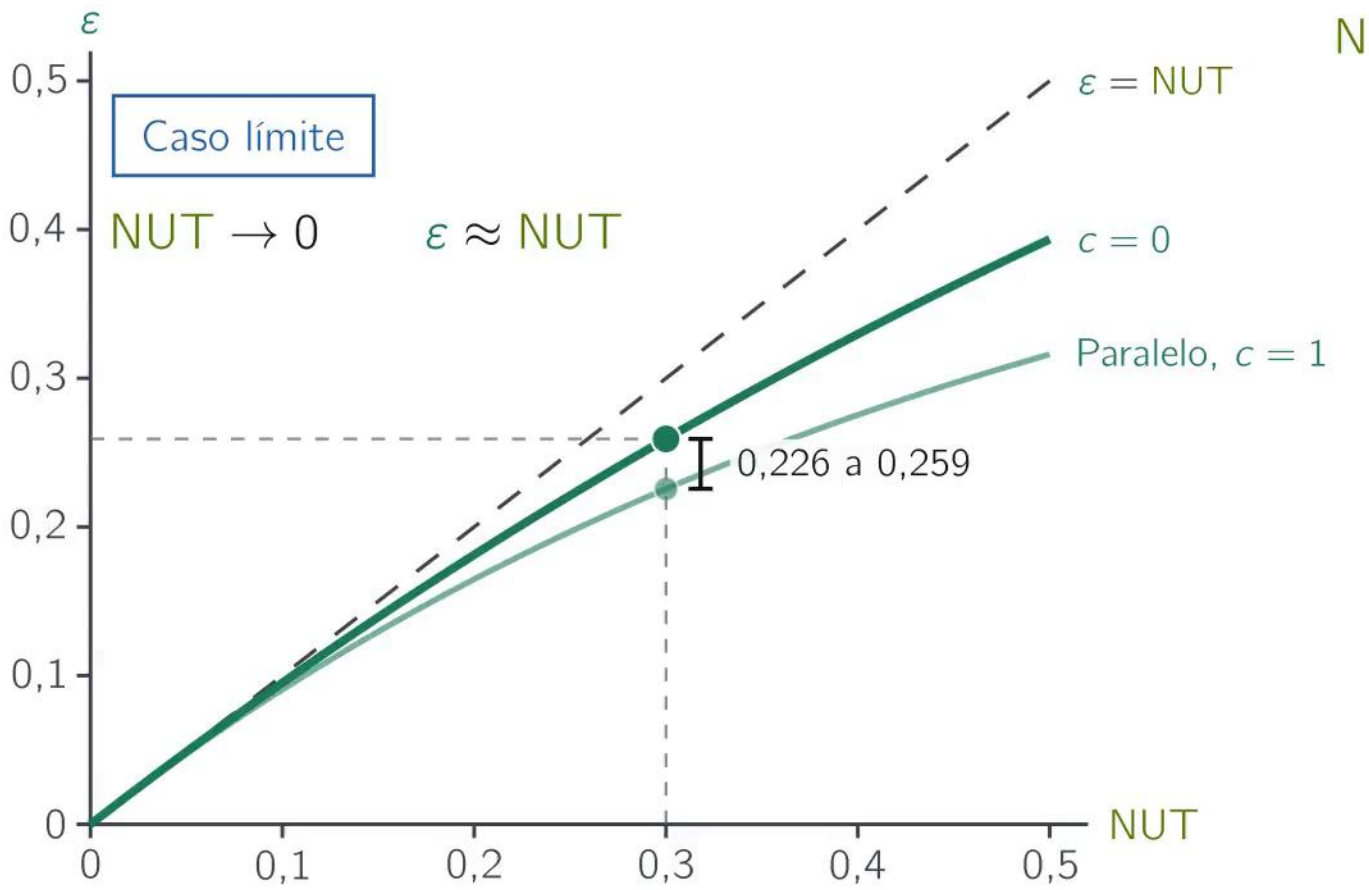
$C_{min} = C_c$





# Efectividad contra NUT

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón
$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia

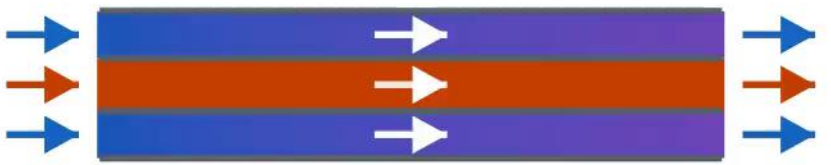


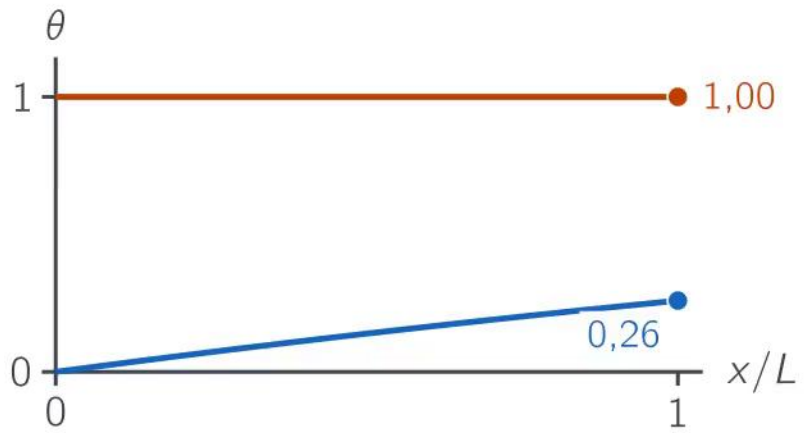

$c = 0{,}00$

$\text{NUT} = 0{,}30$

$\varepsilon = 0{,}26$

$C_{min} = C_c$

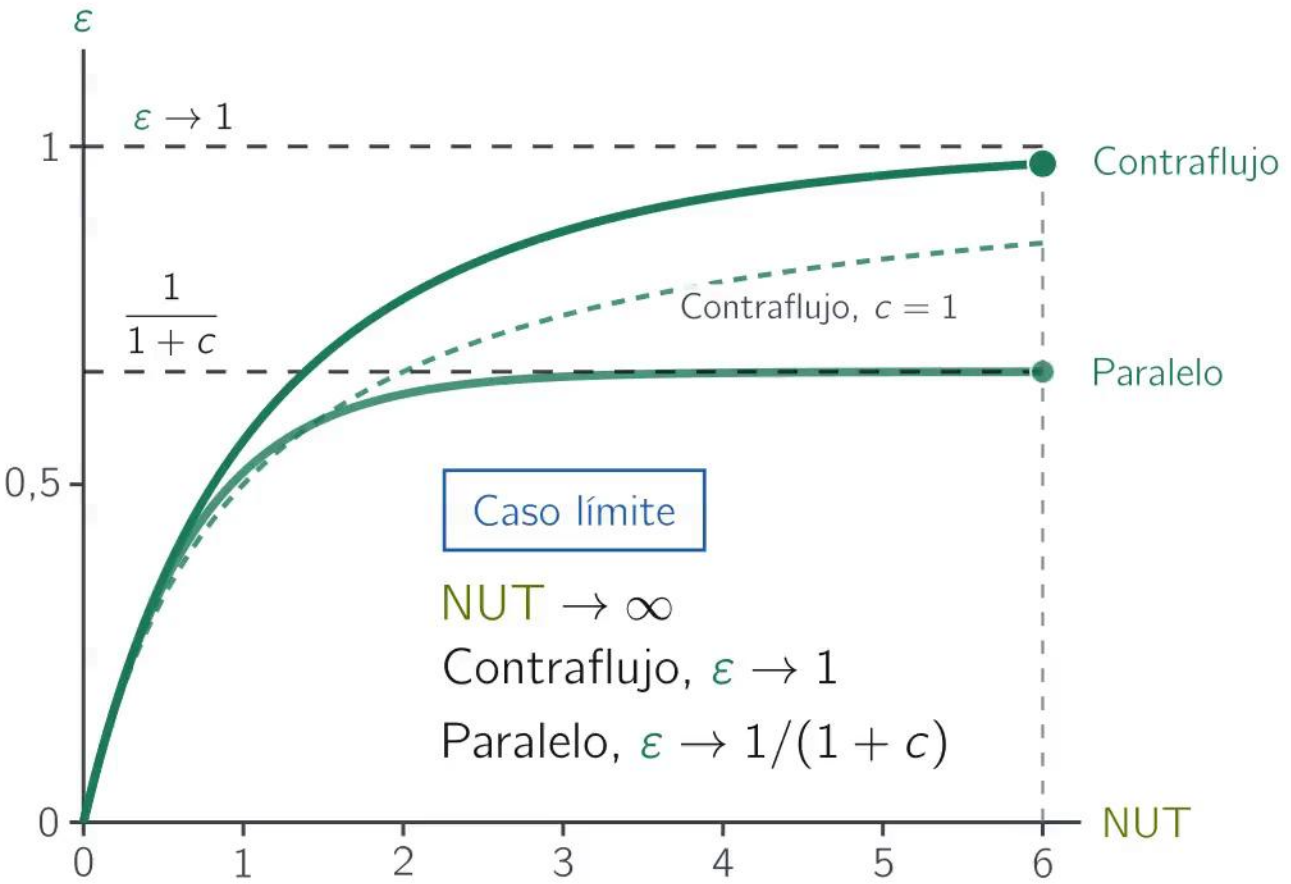




# Efectividad contra NUT

$\varepsilon$ efectividad  NUT unidades de transferencia  $c$ razón

$\dot{Q}_{max}$ máximo  $C_{min}$ menor  $UA_s$ conductancia

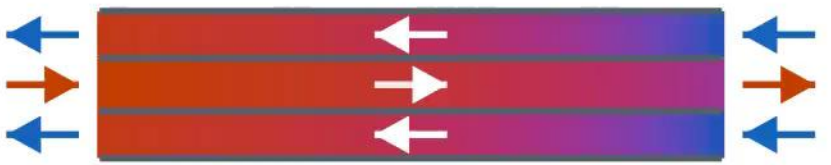


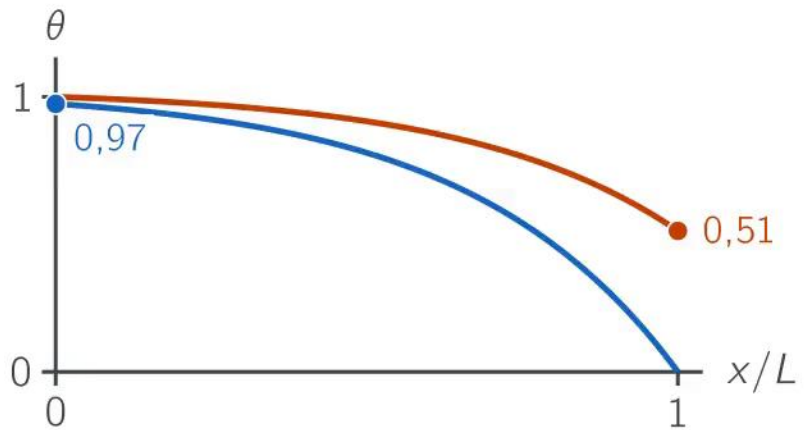

$c = 0{,}50$

$\text{NUT} = 6{,}00$

$\varepsilon = 0{,}97$

$C_{min} = C_c$

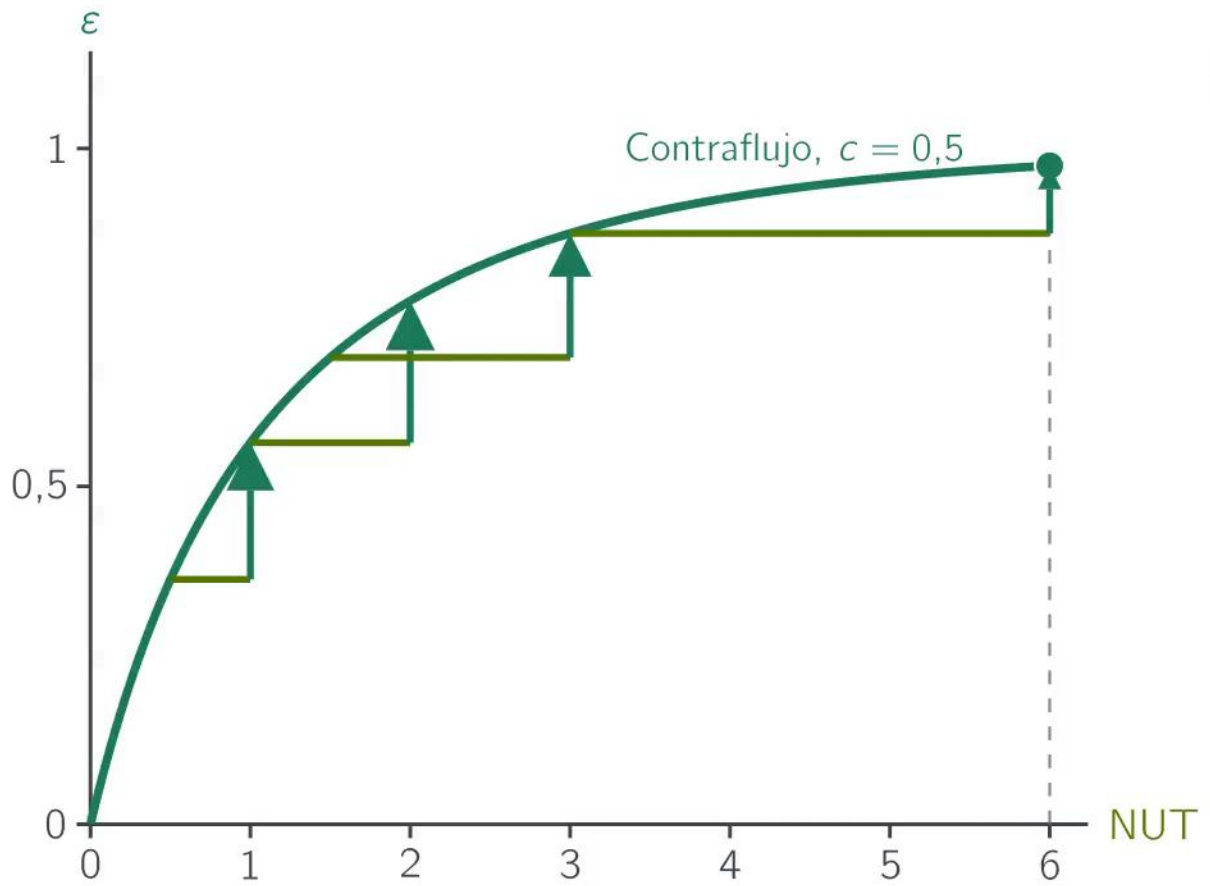




# Duplicar el área

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia




Error frecuente

~~Doble NUT, doble $\dot{Q}$~~

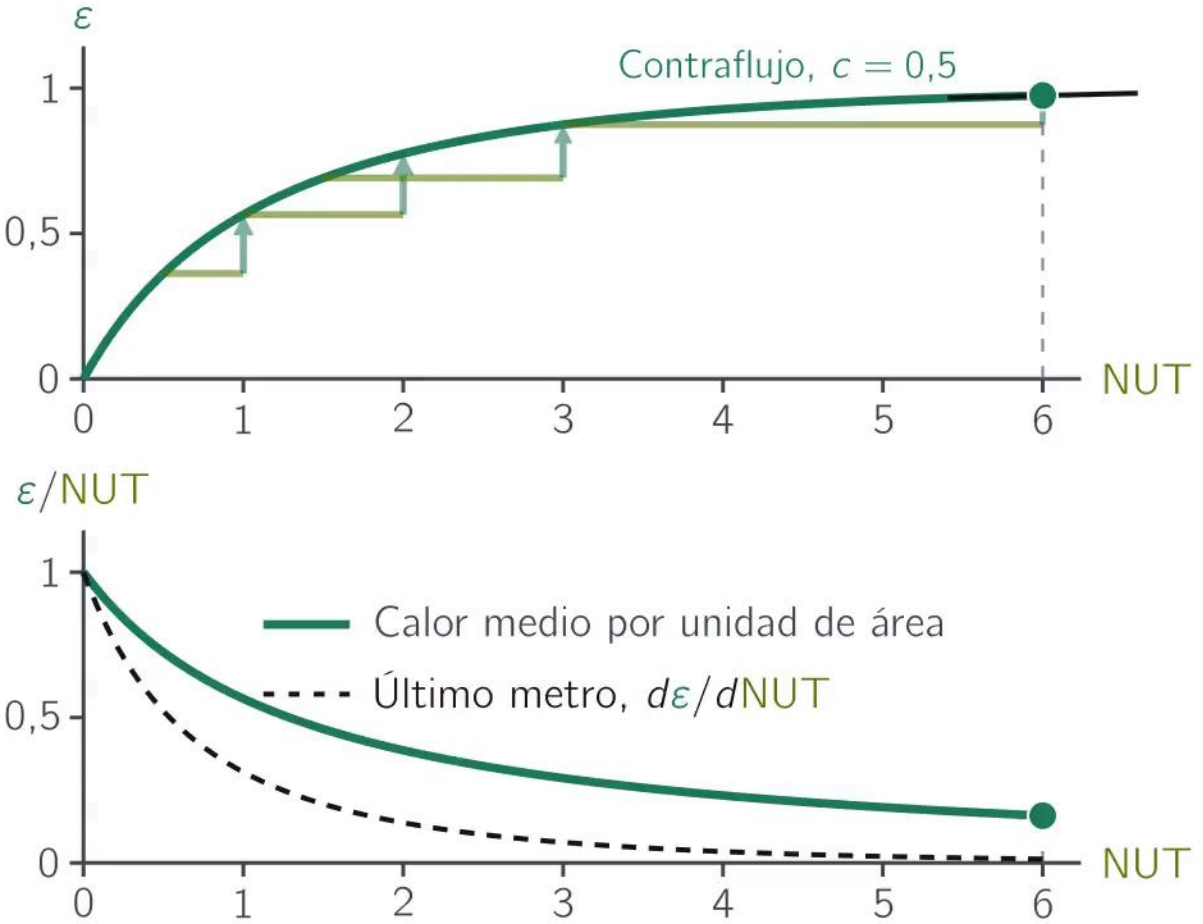
| | | |
|---|---|---|
| $0{,}5 \to 1$ | $\dot{Q}$ | $+\ 56\,\%$ |
| $1 \to 2$ | $\dot{Q}$ | $+\ 37\,\%$ |
| $1{,}5 \to 3$ | $\dot{Q}$ | $+\ 27\,\%$ |
| $3 \to 6$ | $\dot{Q}$ | $+\ 11\,\%$ |

- ☐ Sí
- ■ Menos del doble
- ☐ Más del doble

# Duplicar el área

$\varepsilon$ efectividad NUT unidades de transferencia $c$ razón

$\dot{Q}_{max}$ máximo $C_{min}$ menor $UA_s$ conductancia



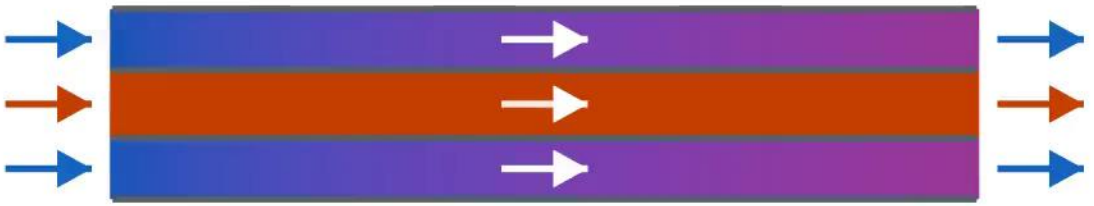

Pregunta de control

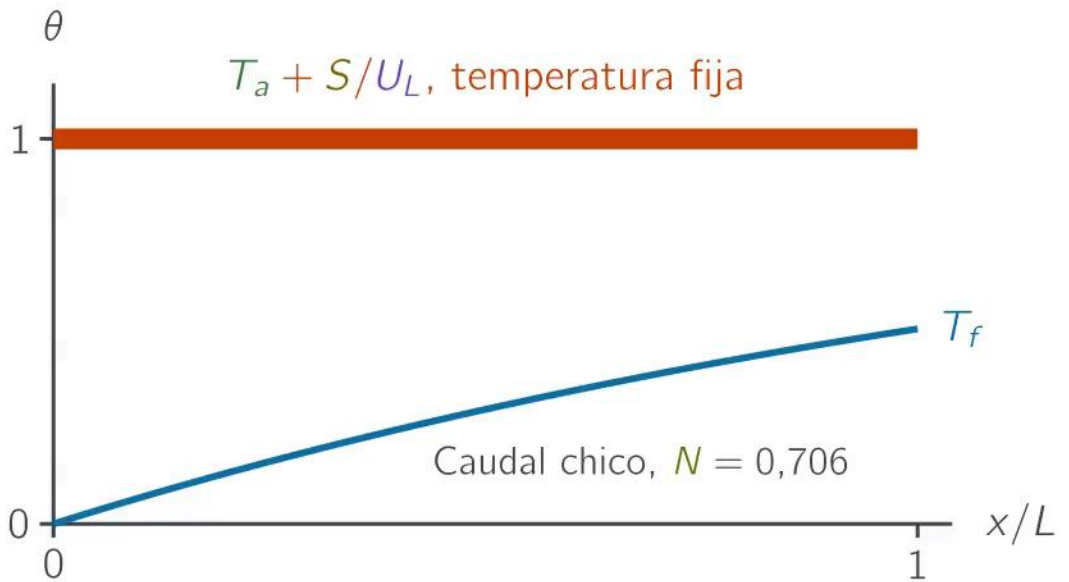
¿Dónde dejarían de agrandar el equipo?

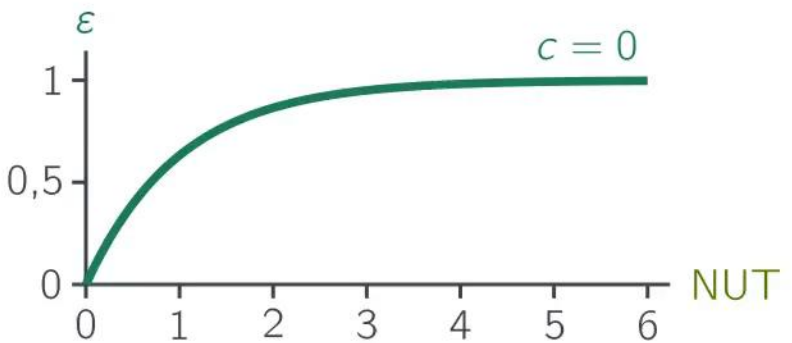
En NUT = 3
Pendiente 0,071, promedio 0,291

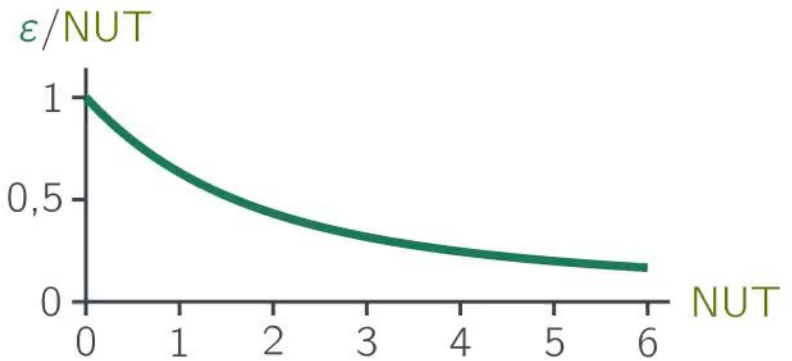
# Colector solar

$S$ absorbida $T_a$ ambiente $U_L$ pérdidas
$N$ NUT colector $F'$ eficiencia del colector $F_R$ remoción



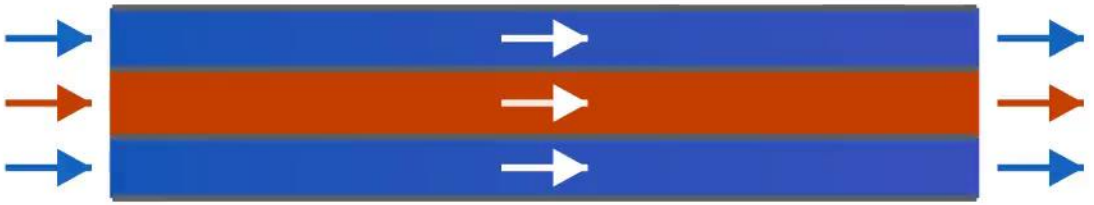


# Colector solar

$S$ absorbida $T_a$ ambiente $U_L$ pérdidas
$N$ NUT colector $F'$ eficiencia del colector $F_R$ remoción

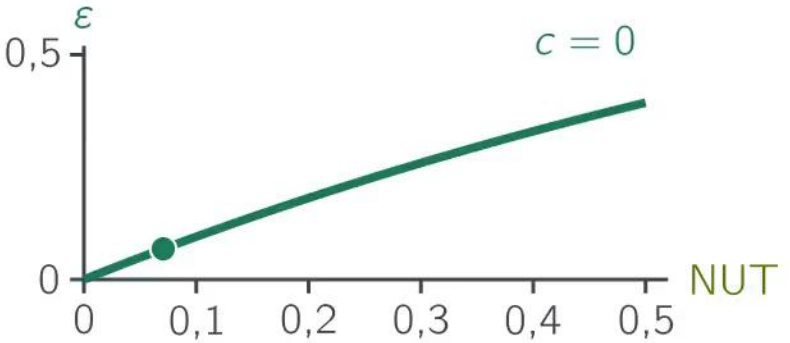


$$\frac{F_R}{F'} = \frac{1 - e^{-N}}{N} = \frac{\varepsilon}{\text{NUT}}$$

$$UA_s \leftrightarrow A_c U_L F'$$

$$C_{\min} \leftrightarrow \dot{m} c_p$$

$$\text{NUT} \leftrightarrow N = \frac{A_c U_L F'}{\dot{m} c_p}$$



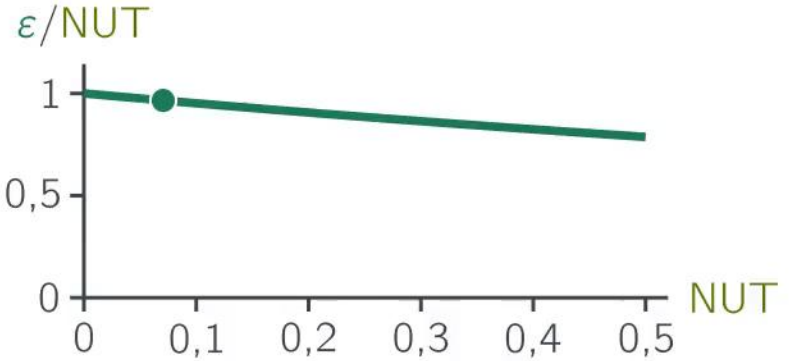

Caudal chico, $N = 0{,}706$

Caudal de la clase 6

$N = 0{,}0706$

$\varepsilon = 0{,}068$

$F_R/F' = 0{,}966$



# Contenidos


1 Evaluar un equipo

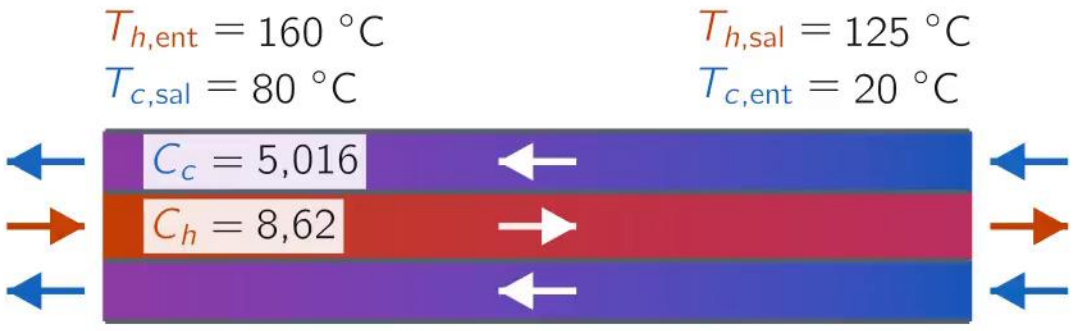
2 Calor máximo y efectividad

3 Curvas $\varepsilon$–NUT

4 Ejemplos y selección

# Geotermia con $\varepsilon$–NUT

$C_h$ caliente $C_c$ fría $c$ razón
$\varepsilon$ efectividad NUT unidades de transferencia $\dot{Q}_{\max}$ máximo




$\dot{m}_h = 2$ kg/s $\quad \dot{m}_c = 1{,}2$ kg/s

$c_{p,h} = 4{,}31$ kJ/(kg · °C) $\quad c_{p,c} = 4{,}18$ kJ/(kg · °C)

$U = 640$ W/(m$^2$ · °C) $\quad D = 1{,}5$ cm $\quad \dot{Q} = 301{,}0$ kW

$$C_h = \dot{m}_h c_{p,h} = 2 \cdot 4{,}31 = 8{,}62 \text{ kW/°C}$$

$$C_c = \dot{m}_c c_{p,c} = 1{,}2 \cdot 4{,}18 = 5{,}016 \text{ kW/°C}$$

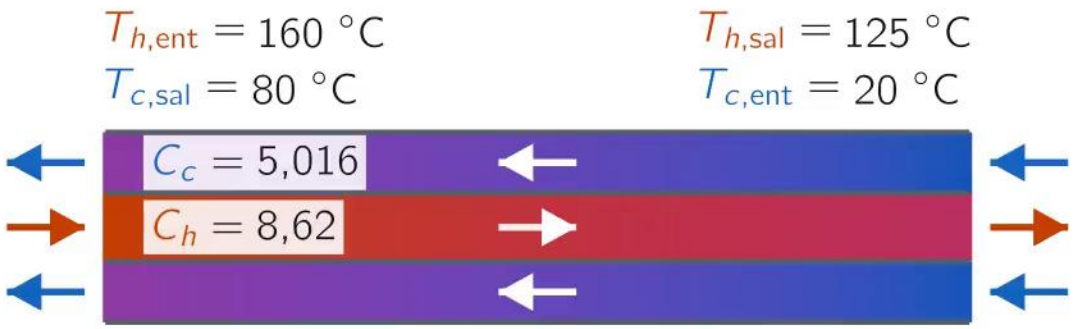
$$C_{\min} = C_c, \quad c = \frac{C_{\min}}{C_{\max}} = \frac{5{,}016}{8{,}62} = 0{,}582$$

$$\dot{Q}_{\max} = C_{\min}(T_{h,\text{ent}} - T_{c,\text{ent}})$$

$$= 5{,}016 \cdot (160 - 20) = 702{,}2 \text{ kW}$$

# Geotermia con $\varepsilon$–NUT

$C_h$ caliente $C_c$ fría $c$ razón
$\varepsilon$ efectividad NUT unidades de transferencia $\dot{Q}_{\max}$ máximo

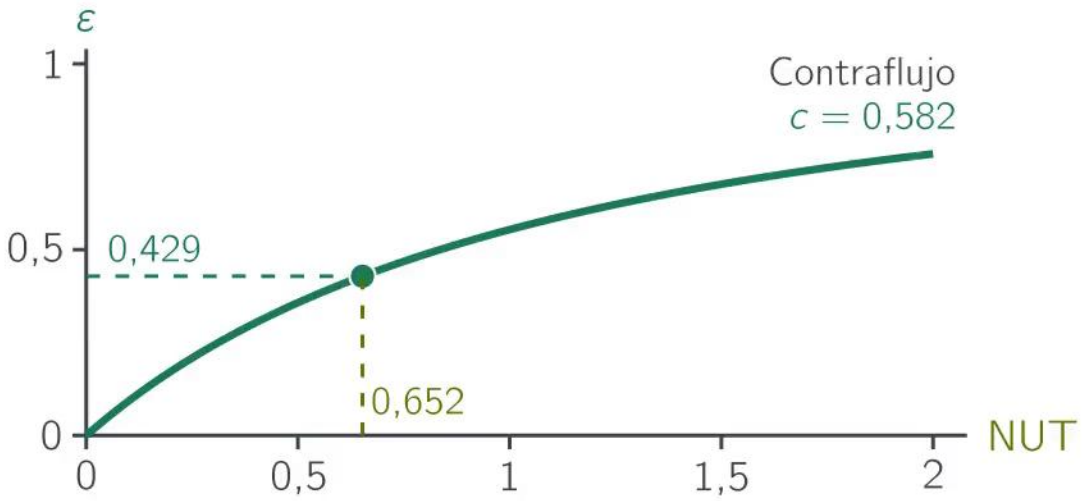



$$c = 0{,}582, \quad \dot{Q}_{\max} = 702{,}2\ \text{kW}$$

$$\varepsilon = \frac{\dot{Q}}{\dot{Q}_{\max}} = \frac{301{,}0}{702{,}2} = 0{,}429$$

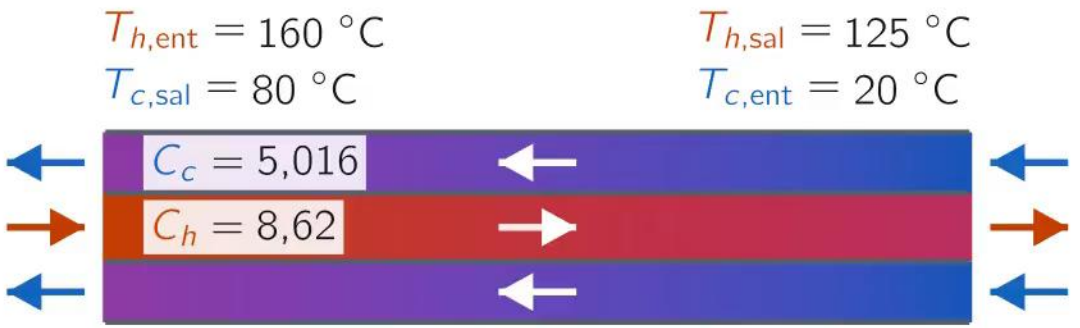
Contraflujo, despejado para NUT

$$\text{NUT} = \frac{1}{1-c} \ln \frac{1-\varepsilon c}{1-\varepsilon} = 0{,}652$$



# Geotermia con $\varepsilon$–NUT

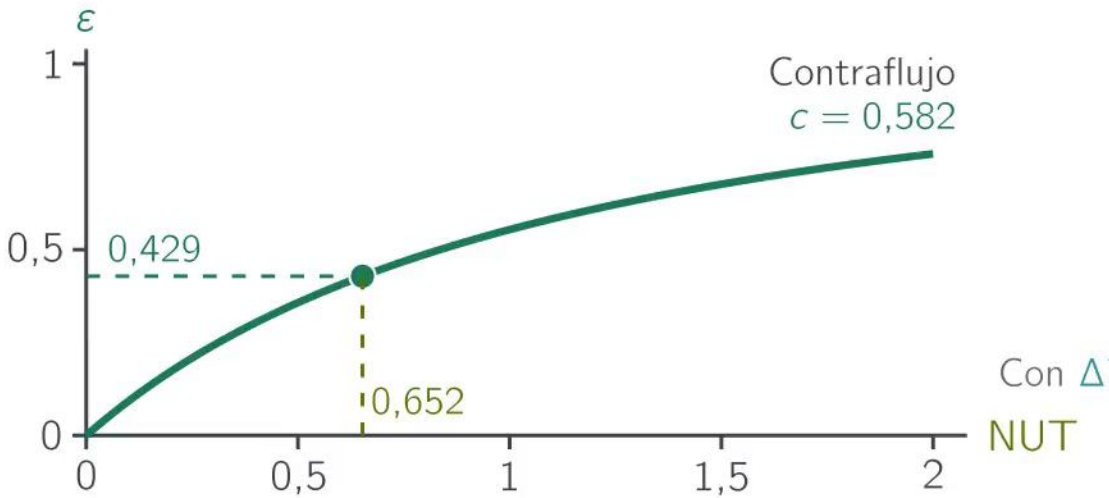
$C_h$ caliente $C_c$ fría $c$ razón
$\varepsilon$ efectividad NUT unidades de transferencia $\dot{Q}_{max}$ máximo




$$c = 0{,}582, \quad \dot{Q}_{max} = 702{,}2\ \text{kW}$$

$$\text{NUT} = \frac{1}{1-c} \ln \frac{1-\varepsilon c}{1-\varepsilon} = 0{,}652$$

$$A_s = \frac{\text{NUT}\, C_{min}}{U} = \frac{0{,}652 \cdot 5{,}016 \cdot 10^3}{640} = 5{,}11\ \text{m}^2$$



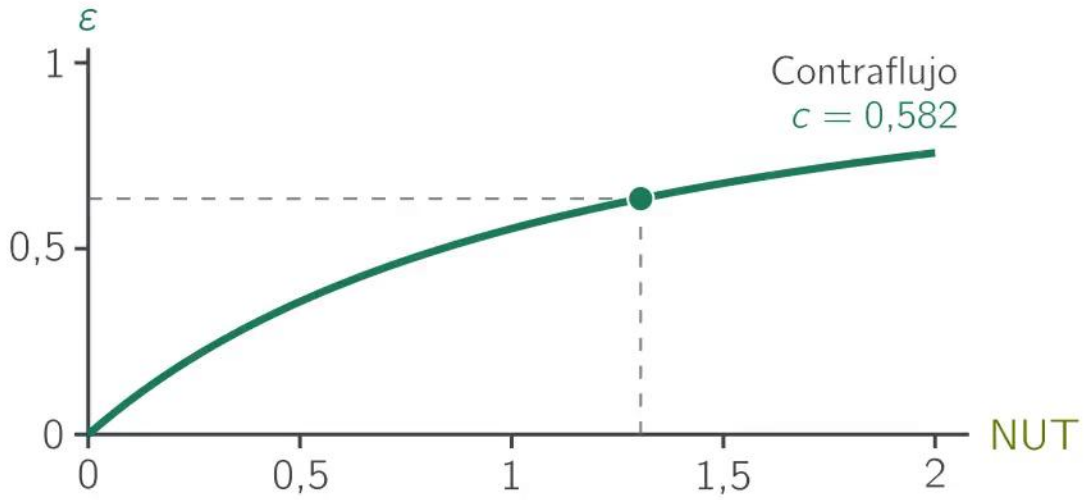

$$L = \frac{A_s}{\pi D} = 108{,}5\ \text{m}$$

Con $\Delta T_{ml}$, sesión 2 $L = 108{,}5$ m

# Geotermia con $\varepsilon$–NUT

$C_h$ caliente $C_c$ fría $c$ razón
$\varepsilon$ efectividad NUT unidades de transferencia $\dot{Q}_{\max}$ máximo




$\text{NUT} = 1{,}305$

$\varepsilon = 0{,}634$

$\dot{Q} = 0{,}634 \cdot 702{,}2 = 445\ \text{kW}, \quad \dfrac{445}{301{,}0} = 1{,}48$

$T_{c,\text{sal}} = 20 + 0{,}634 \cdot (160 - 20) = 108{,}8\ ^\circ\text{C}$

$T_{h,\text{sal}} = 160 - 0{,}582 \cdot 0{,}634 \cdot (160 - 20) = 108{,}3\ ^\circ\text{C}$

Pregunta de control

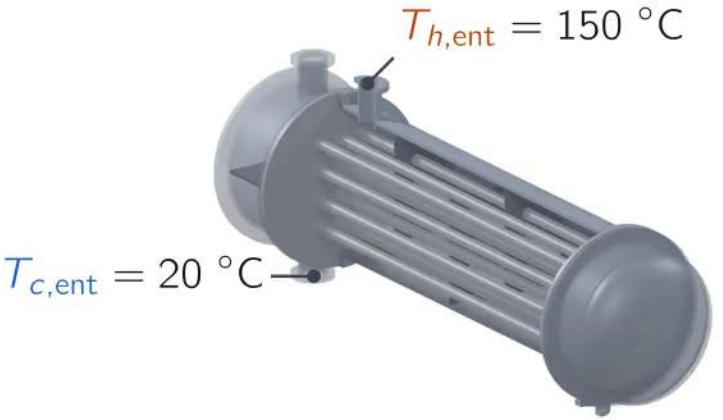
Si duplico el intercambiador, ¿duplico el calor?

- ☐ Sí
- ■ Menos del doble
- ☐ Más del doble

# Enfriador de aceite

$C_h$ caliente $C_c$ fría $c$ razón
$\varepsilon$ efectividad NUT unidades de transferencia $\dot{Q}_{max}$ máximo

1. $C_h$ y $C_c$
2. $C_{min}$, $c$ y $\dot{Q}_{max}$
3. $\text{NUT} = UA_s/C_{min}$
4. $\varepsilon$ con fórmula o gráfico
5. $\dot{Q} = \varepsilon \dot{Q}_{max}$ y salidas por balance




Aceite en el casco y agua en los tubos

$\dot{m}_h = 0{,}3$ kg/s, $c_{p,h} = 2{,}13$ kJ/(kg · °C)

$\dot{m}_c = 0{,}2$ kg/s, $c_{p,c} = 4{,}18$ kJ/(kg · °C)

$D = 1{,}4$ cm

8 pasos de 5 m por los tubos

$U = 310$ W/(m$^2$ · °C)

$$C_h = 0{,}3 \cdot 2{,}13 = 0{,}639 \text{ kW/°C}$$

$$C_c = 0{,}2 \cdot 4{,}18 = 0{,}836 \text{ kW/°C}$$

$$C_{min} = C_h, \quad c = \frac{0{,}639}{0{,}836} = 0{,}764$$

$$\dot{Q}_{max} = 0{,}639 \cdot (150 - 20) = 83{,}1 \text{ kW}$$

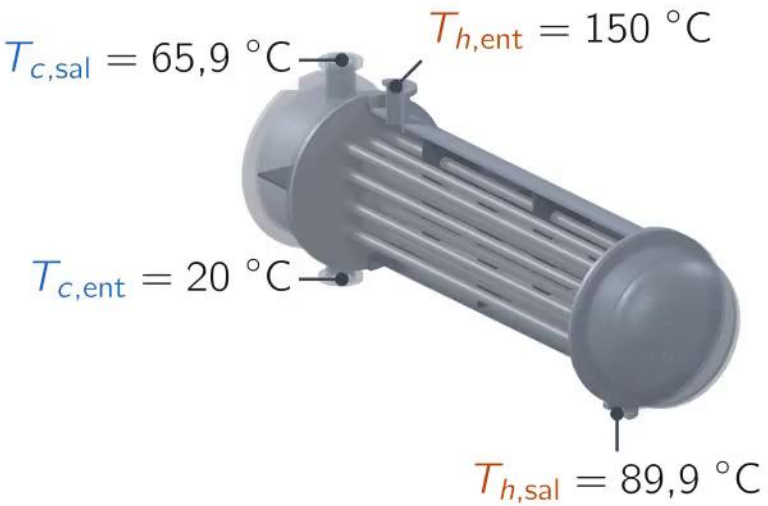
$$A_s = 8\pi DL = 8 \cdot \pi \cdot 0{,}014 \cdot 5 = 1{,}759 \text{ m}^2$$

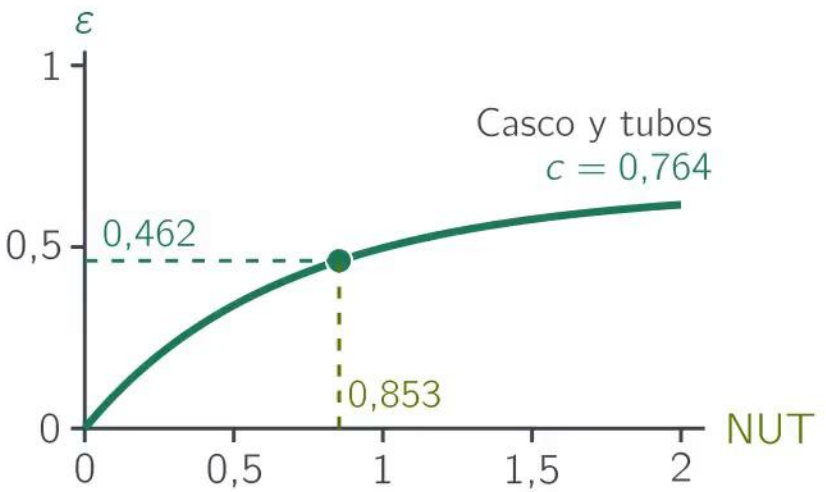
$$\text{NUT} = \frac{UA_s}{C_{min}} = \frac{310 \cdot 1{,}759}{0{,}639 \cdot 10^3} = 0{,}853$$

# Enfriador de aceite

$C_h$ caliente $C_c$ fría $c$ razón
$\varepsilon$ efectividad NUT unidades de transferencia $\dot{Q}_{max}$ máximo

1. $C_h$ y $C_c$
2. $C_{min}$, $c$ y $\dot{Q}_{max}$
3. $\text{NUT} = UA_s/C_{min}$
4. $\varepsilon$ con fórmula o gráfico
5. $\dot{Q} = \varepsilon \dot{Q}_{max}$ y salidas por balance




$$C_{min} = C_h, \quad c = \frac{0{,}639}{0{,}836} = 0{,}764$$

$$\dot{Q}_{max} = 0{,}639 \cdot (150 - 20) = 83{,}1 \text{ kW}$$

$$A_s = 8\pi DL = 8 \cdot \pi \cdot 0{,}014 \cdot 5 = 1{,}759 \text{ m}^2$$

$$\text{NUT} = \frac{UA_s}{C_{min}} = \frac{310 \cdot 1{,}759}{0{,}639 \cdot 10^3} = 0{,}853$$

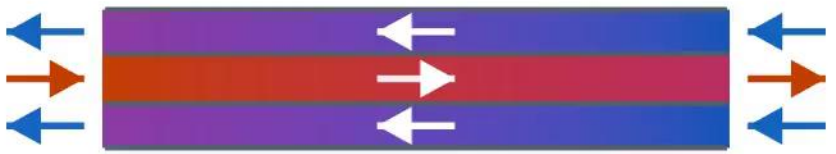
$\varepsilon = 0{,}462$ Casco y tubos, tabla 11-4

$$\dot{Q} = \varepsilon \dot{Q}_{max} = 0{,}462 \cdot 83{,}1 = 38{,}4 \text{ kW}$$

$$T_{c,sal} = 20 + \frac{38{,}4}{0{,}836} = 65{,}9\ °C$$

$$T_{h,sal} = 150 - \frac{38{,}4}{0{,}639} = 89{,}9\ °C$$

# Selección de intercambiadores



Salidas conocidas

$\Delta T_{ml}$

Se busca el tamaño

Salidas desconocidas

$\varepsilon$–NUT

Se evalúa un equipo dado

Los dos dan el mismo equipo

ejemplo geotérmico $L = 108{,}5$ m

# Selección de intercambiadores

U incierto — Se compra un equipo algo más grande

1. **Calor**
   Entregar $\dot{Q}$ con las temperaturas pedidas
2. **Costo**
   De compra y de operación
3. **Potencia de bombeo**
   Crece con el cubo del caudal $\dot{V}$ en flujo turbulento
4. **Tamaño y peso**
   Espacio disponible en la planta
5. **Tipo y materiales**
   Presión, corrosión e incrustación

Potencia de bombeo

$$\dot{W}_{\text{bomba}} = \frac{\Delta P \dot{V}}{\eta_{\text{bomba}}}$$

$$\Delta P \propto \dot{V}^2 \Rightarrow \dot{W}_{\text{bomba}} \propto \dot{V}^3$$

# Notación

| | |
|---|---|
| $T_h$, $T_c$ | Caliente y frío, ent y sal, °C |
| $U$, $U_i$, $U_o$ | Coeficiente total, W/(m$^2$ · °C) |
| $A_s$, $A_i$, $A_o$ | Área de transferencia, m$^2$ |
| $R_i$, $R_{\text{pared}}$, $R_o$ | Resistencias, °C/W |
| $R_f$ | Incrustación, m$^2$ · °C/W |
| $\Delta T_1$, $\Delta T_2$ | Diferencias en los extremos, °C |
| $\Delta T_{ml}$ | Media logarítmica, °C |
| $F$ | Factor de corrección |
| $P$, $R_F$ | Diagramas de $F$, $R_F$ puede pasar de 1 |

| | |
|---|---|
| $C_h$, $C_c = \dot{m}c_p$ | Capacidad calorífica, W/°C |
| $C_{\min}$, $C_{\max}$ | Menor y mayor de las dos |
| $c = C_{\min}/C_{\max}$ | Razón de capacidades, 0 a 1 |
| $\Delta T_{\max}$ | $T_{h,\text{ent}} - T_{c,\text{ent}}$, mayor diferencia |
| $\dot{Q}_{\max} = C_{\min}\Delta T_{\max}$ | Calor máximo, W |
| $\varepsilon = \dot{Q}/\dot{Q}_{\max}$ | Efectividad |
| $\text{NUT} = UA_s/C_{\min}$ | Unidades de transferencia |
| $\theta$ | 0 en $T_{c,\text{ent}}$ y 1 en $T_{h,\text{ent}}$ |
| $N$, $F_R$, $F'$ | Colector, $N$ hace de NUT |
| $S$, $T_a$, $U_L$, $A_c$ | Datos del colector, clase 6 |

> $F$ no es la eficiencia de aleta. $R_F$ no es $c$ ni $R_f$.

# Hipótesis y próxima clase

## Hipótesis de hoy

- Régimen estacionario, equipo aislado del exterior
- $U$ y $c_p$ constantes, cambio de fase solo a temperatura fija ($c \to 0$)
- Sin conducción a lo largo del tubo

## Próxima clase, combustión

- ¿Cuánta energía entrega un kilo de gas y cuánto aire necesita?
- El intercambiador vuelve como precalentador de aire
- Lado gas con $h$ bajo e incrustación

